


```
****A  START  JOB  955  HANOIASM  Hanoi ASM            ROOM       8.50.50 PM 01 NOV 26  PRINTER1  SYS TK4-  JOB  955  START  A****
****A  START  JOB  955  HANOIASM  Hanoi ASM            ROOM       8.50.50 PM 01 NOV 26  PRINTER1  SYS TK4-  JOB  955  START  A****
****A  START  JOB  955  HANOIASM  Hanoi ASM            ROOM       8.50.50 PM 01 NOV 26  PRINTER1  SYS TK4-  JOB  955  START  A****
```

J E S 2 J O B L O G

```
20.50.49 JOB  955  $HASP373 HANOIASM STARTED - INIT  1 - CLASS A - SYS TK4-
20.50.49 JOB  955  IEF403I HANOIASM - STARTED - TIME=20.50.49
20.50.49 JOB  955  IEFACTRT - Stepname  Procstep  Program   Retcode
20.50.49 JOB  955  HANOIASM   HANOIREC  ASM       IFOX00    RC= 0000
20.50.49 JOB  955  HANOIASM   HANOIITR  ASM       IFOX00    RC= 0000
20.50.50 JOB  955  HANOIASM   HANOIASM  ASM       IFOX00    RC= 0000
20.50.50 JOB  955  HANOIASM   HANOIASM  LKED      IEWL      RC= 0000
20.50.50 JOB  955  +HANOIASM... Begin execution
20.50.50 JOB  955  +HANOIASM... calling HANOIITR
20.50.50 JOB  955  +HANOIITR... start HANOIITR
20.50.50 JOB  955  +HANOIITR... end HANOIITR
20.50.50 JOB  955  +HANOIASM... back from HANOIITR
20.50.50 JOB  955  +HANOIASM... Normal end of program
20.50.50 JOB  955  HANOIASM   HANOIASM  GO        PGM=*.DD  RC= 0000
20.50.50 JOB  955  IEF404I HANOIASM - ENDED - TIME=20.50.50
20.50.50 JOB  955  $HASP395 HANOIASM ENDED


------ JES2 JOB STATISTICS ------


 01 NOV 26 JOB EXECUTION DATE


       855 CARDS READ


     1,955 SYSOUT PRINT RECORDS


         0 SYSOUT PUNCH RECORDS


      0.02 MINUTES EXECUTION TIME
```

```
1     //HANOIASM JOB (MBA),                                                JOB  955
      //             'Hanoi ASM',                                          00000201
      //             CLASS=A,                                              00000300
      //             MSGCLASS=A,                                           00000400
      //             NOTIFY=HERC01,                                        00000501
      //             TIME=1440,                                            00000600
      //             MSGLEVEL=(1,1),                                       00000700
      //           USER=HERC01,PASSWORD=           GENERATED BY GDL
      ************************************************************------- 00000801
      *** Name: HERC01.MYSTUFF.JCL(HANOIASM)                              00000901
      ***                                                                 00001001
      *** Description: Towers of Hanoi. Please refer to                   00001121
      ***   https://en.wikipedia.org/wiki/Tower_of_Hanoi                  00001221
      ***   The job contains the section HANOIASM as entry point (main    00001321
      ***   program) and two variants of handling the movement of the     00001421
      ***   slices: a recursive one (section HANOIREC) and an iterative one 00001521
      ***   (section HANOIITR). The sections are assembled, linked and    00001621
      ***   executed                                                      00001721
      *** Input: Number of slices slices (1<=slices<=20), source stack from 00001821
      ***   (0<=from<=2), destination stack to (0<=to<=2) (from<>to) are   00001921
      ***   entered in //GO.HANOIIN DD                                    00002021
      *** Output: Print output to //GO.HANOIOUT DD that shows the steps with 00002121
      ***   information on which slice is moved from/to which stack       00002221
      ************************************************************------- 00002301
      ***                                                                 00002400
      ***---------------------------------------------------------------- 00002501
2     //HANOIREC EXEC ASMFC,PARM.ASM=(LOAD,NODECK),MAC1='SYS2.MACLIB'      00002601
3     XXASMFC    PROC  MAC='SYS1.MACLIB',MAC1='SYS1.MACLIB',              00000100
      XX         MAC2='SYS1.MACLIB',MAC3='SYS1.MACLIB',SOUT='*'           00000200
4     XXASM      EXEC  PGM=IFOX00,REGION=128K                             00000300
5     XXSYSLIB   DD    DSN=&MAC,DISP=SHR                                  00000400
6     XX         DD    DSN=&MAC1,DISP=SHR                                 00000500
7     XX         DD    DSN=&MAC2,DISP=SHR                                 00000600
8     XX         DD    DSN=&MAC3,DISP=SHR                                 00000700
9     XXSYSUT1   DD    DSN=&&SYSUT1,UNIT=SYSSQ,SPACE=(1700,(600,100)),    00000800
      XX             SEP=(SYSLIB)                                         00000900
10    XXSYSUT2   DD    DSN=&&SYSUT2,UNIT=SYSSQ,SPACE=(1700,(300,50)),     00001000
      XX             SEP=(SYSLIB,SYSUT1)                                  00001100
11    XXSYSUT3   DD    DSN=&&SYSUT3,UNIT=SYSSQ,SPACE=(1700,(300,50))      00001200
12    XXSYSPRINT DD    SYSOUT=&SOUT,DCB=BLKSIZE=1089                      00001300
13    XXSYSPUNCH DD    SYSOUT=B                                           00001400
14    //ASM.SYSGO DD DSNAME=&&OBJSET,UNIT=SYSSQ,                         X00002701
      //             SPACE=(80,(200,50)),                                X00002801
      //             DISP=(MOD,PASS)                                      00002901
15    //ASM.SYSIN DD *                                                   00003001
      ***---------------------------------------------------------------- 00023401
16    //HANOIITR EXEC ASMFC,PARM.ASM=(LOAD,NODECK),MAC1='SYS2.MACLIB'      00023501
17    XXASMFC    PROC  MAC='SYS1.MACLIB',MAC1='SYS1.MACLIB',              00000100
      XX         MAC2='SYS1.MACLIB',MAC3='SYS1.MACLIB',SOUT='*'           00000200
18    XXASM      EXEC  PGM=IFOX00,REGION=128K                             00000300
19    XXSYSLIB   DD    DSN=&MAC,DISP=SHR                                  00000400
20    XX         DD    DSN=&MAC1,DISP=SHR                                 00000500
21    XX         DD    DSN=&MAC2,DISP=SHR                                 00000600
22    XX         DD    DSN=&MAC3,DISP=SHR                                 00000700
23    XXSYSUT1   DD    DSN=&&SYSUT1,UNIT=SYSSQ,SPACE=(1700,(600,100)),    00000800
      XX             SEP=(SYSLIB)                                         00000900
24    XXSYSUT2   DD    DSN=&&SYSUT2,UNIT=SYSSQ,SPACE=(1700,(300,50)),     00001000
      XX             SEP=(SYSLIB,SYSUT1)                                  00001100
25    XXSYSUT3   DD    DSN=&&SYSUT3,UNIT=SYSSQ,SPACE=(1700,(300,50))      00001200
26    XXSYSPRINT DD    SYSOUT=&SOUT,DCB=BLKSIZE=1089                      00001300
27    XXSYSPUNCH DD    SYSOUT=B                                           00001400
28    //ASM.SYSGO DD DSNAME=&&OBJSET,UNIT=SYSSQ,                         X00023601
      //             SPACE=(80,(200,50)),                                X00023701
```

```
      //             DISP=(MOD,PASS)                                          00023801
29    //ASM.SYSIN DD *                                                       00023901
      ***--------------------------------------------------------------------00061201
30    //HANOIASM EXEC ASMFCLG,PARM.ASM=(OBJ,NODECK),MAC1='SYS2.MACLIB',      X00061301
      //             REGION.GO=128K                                          00061401
31    XXASMFCLG  PROC MAC='SYS1.MACLIB',MAC1='SYS1.MACLIB',                 00000100
      XX         MAC2='SYS1.MACLIB',MAC3='SYS1.MACLIB',SOUT='*'             00000200
32    XXASM      EXEC PGM=IFOX00,PARM=OBJ,REGION=128K                       00000300
33    XXSYSLIB   DD   DSN=&MAC,DISP=SHR                                     00000400
34    XX         DD   DSN=&MAC1,DISP=SHR                                    00000500
35    XX         DD   DSN=&MAC2,DISP=SHR                                    00000600
36    XX         DD   DSN=&MAC3,DISP=SHR                                    00000700
37    XXSYSUT1   DD   DSN=&&SYSUT1,UNIT=SYSSQ,SPACE=(1700,(600,100)),       00000800
      XX             SEP=(SYSLIB)                                           00000900
38    XXSYSUT2   DD   DSN=&&SYSUT2,UNIT=SYSSQ,SPACE=(1700,(300,50)),        00001000
      XX             SEP=(SYSLIB,SYSUT1)                                    00001100
39    XXSYSUT3   DD   DSN=&&SYSUT3,UNIT=SYSSQ,SPACE=(1700,(300,50))         00001200
40    XXSYSPRINT DD   SYSOUT=&SOUT,DCB=BLKSIZE=1089                         00001300
41    XXSYSPUNCH DD   SYSOUT=B                                              00001400
42    XXSYSGO    DD   DSN=&&OBJSET,UNIT=SYSSQ,SPACE=(80,(200,50)),         00001500
      XX             DISP=(MOD,PASS)                                        00001600
43    //ASM.SYSIN DD *                                                      00061501
      ***--------------------------------------------------------------------00084601
44    XXLKED     EXEC PGM=IEWL,PARM=(XREF,LET,LIST,NCAL),REGION=128K,       00001700
      XX             COND=(8,LT,ASM)                                        00001800
45    XXSYSLIN   DD   DSN=&&OBJSET,DISP=(OLD,DELETE)                        00001900
46    XX         DD   DDNAME=SYSIN                                          00002000
47    XXSYSLMOD  DD   DSN=&&GOSET(GO),UNIT=SYSDA,SPACE=(1024,(50,20,1)),    00002100
      XX             DISP=(MOD,PASS)                                        00002200
48    XXSYSUT1   DD   DSN=&&SYSUT1,UNIT=(SYSDA,SEP=(SYSLIN,SYSLMOD)),      00002300
      XX             SPACE=(1024,(50,20))                                   00002400
49    XXSYSPRINT DD   SYSOUT=&SOUT                                          00002500
50    //LKED.SYSIN  DD *                                                    00084701
51    //SYSIN     DD *                 GENERATED STATEMENT
52    XXGO       EXEC PGM=*.LKED.SYSLMOD,COND=((8,LT,ASM),(4,LT,LKED))      00002600
53    //GO.HANOIIN  DD *                                                    00085101
54    //GO.HANOIOUT DD SYSOUT=*,DCB=(RECFM=FBA,LRECL=133,BLKSIZE=1330)      00085401
```

```
 STMT NO. MESSAGE
-
    5     IEF653I SUBSTITUTION JCL - DSN=SYS1.MACLIB,DISP=SHR
    6     IEF653I SUBSTITUTION JCL - DSN=SYS2.MACLIB,DISP=SHR
    7     IEF653I SUBSTITUTION JCL - DSN=SYS1.MACLIB,DISP=SHR
    8     IEF653I SUBSTITUTION JCL - DSN=SYS1.MACLIB,DISP=SHR
   12     IEF653I SUBSTITUTION JCL - SYSOUT=*,DCB=BLKSIZE=1089
   19     IEF653I SUBSTITUTION JCL - DSN=SYS1.MACLIB,DISP=SHR
   20     IEF653I SUBSTITUTION JCL - DSN=SYS2.MACLIB,DISP=SHR
   21     IEF653I SUBSTITUTION JCL - DSN=SYS1.MACLIB,DISP=SHR
   22     IEF653I SUBSTITUTION JCL - DSN=SYS1.MACLIB,DISP=SHR
   26     IEF653I SUBSTITUTION JCL - SYSOUT=*,DCB=BLKSIZE=1089
   33     IEF653I SUBSTITUTION JCL - DSN=SYS1.MACLIB,DISP=SHR
   34     IEF653I SUBSTITUTION JCL - DSN=SYS2.MACLIB,DISP=SHR
   35     IEF653I SUBSTITUTION JCL - DSN=SYS1.MACLIB,DISP=SHR
   36     IEF653I SUBSTITUTION JCL - DSN=SYS1.MACLIB,DISP=SHR
   40     IEF653I SUBSTITUTION JCL - SYSOUT=*,DCB=BLKSIZE=1089
   49     IEF653I SUBSTITUTION JCL - SYSOUT=*
IEF236I ALLOC. FOR HANOIASM ASM HANOIREC
IEF237I 148  ALLOCATED TO SYSLIB
IEF237I 148  ALLOCATED TO
IEF237I 148  ALLOCATED TO
IEF237I 148  ALLOCATED TO
IEF237I 190  ALLOCATED TO SYSUT1
IEF237I 170  ALLOCATED TO SYSUT2
IEF237I 180  ALLOCATED TO SYSUT3
IEF237I JES2 ALLOCATED TO SYSPRINT
IEF237I JES2 ALLOCATED TO SYSPUNCH
IEF237I 140  ALLOCATED TO SYSGO
IEF237I JES2 ALLOCATED TO SYSIN
IEF142I HANOIASM ASM HANOIREC - STEP WAS EXECUTED - COND CODE 0000
IEF285I   SYS1.MACLIB                                KEPT          *-------25
IEF285I   VOL SER NOS= MVSRES.
IEF285I   SYS2.MACLIB                                KEPT          *--------0
IEF285I   VOL SER NOS= MVSRES.
IEF285I   SYS1.MACLIB                                KEPT          *--------0
IEF285I   VOL SER NOS= MVSRES.
IEF285I   SYS1.MACLIB                                KEPT          *--------0
IEF285I   VOL SER NOS= MVSRES.
IEF285I   SYS26305.T205049.RA000.HANOIASM.SYSUT1      DELETED       *------120
IEF285I   VOL SER NOS= WORK03.
IEF285I   SYS26305.T205049.RA000.HANOIASM.SYSUT2      DELETED       *-------20
IEF285I   VOL SER NOS= WORK01.
IEF285I   SYS26305.T205049.RA000.HANOIASM.SYSUT3      DELETED       *--------8
IEF285I   VOL SER NOS= WORK02.
IEF285I   JES2.JOB00955.SO0107                       SYSOUT
IEF285I   JES2.JOB00955.SO0108                       SYSOUT
IEF285I   SYS26305.T205049.RA000.HANOIASM.OBJSET      PASSED        *-------22
IEF285I   VOL SER NOS= WORK00.
IEF285I   JES2.JOB00955.SI0101                       SYSIN
IEF373I STEP /ASM     / START 26305.2050
IEF374I STEP /ASM     / STOP  26305.2050 CPU    0MIN 00.22SEC SRB    0MIN 00.02SEC VIRT   128K SYS   172K
*********************************************************************************************************
*     1. Jobstep of job: HANOIASM    Stepname: ASM         Program name: IFOX00     Executed on 01.11.26 from 20.50.49 to 20.50.49 *
*         elapsed time  00:00:00,35                 CPU-Identifier:  TK4-           Page-in:      0                                *
*             CPU time  00:00:00,24           Virtual Storage used:    128K         Page-out:     0                                *
*           corr. CPU:  00:00:00,24   CPU time has been corrected by  1 / 1,0  multiplier                                          *
*                                                                                                                                   *
*     I/O Operation                                                                                                                 *
*     Number of records read via DD * or DD DATA:    202                                                                            *
*     148......25 148.......0 148.......0 148.......0 190.....120 170......20 180.......8 DMY.......0 DMY.......0 140......22      *
*     DMY.......0                                                                                                                   *
*                                                                                                                                   *
```

```
*                                    Charge for step (w/o SYSOUT):          0,40                                                *
********************************************************************************************************************************
IEF236I ALLOC. FOR HANOIASM ASM HANOIITR
IEF237I 148  ALLOCATED TO SYSLIB
IEF237I 148  ALLOCATED TO
IEF237I 148  ALLOCATED TO
IEF237I 148  ALLOCATED TO
IEF237I 190  ALLOCATED TO SYSUT1
IEF237I 180  ALLOCATED TO SYSUT2
IEF237I 170  ALLOCATED TO SYSUT3
IEF237I JES2 ALLOCATED TO SYSPRINT
IEF237I JES2 ALLOCATED TO SYSPUNCH
IEF237I 140  ALLOCATED TO SYSGO
IEF237I JES2 ALLOCATED TO SYSIN
IEF142I HANOIASM ASM HANOIITR - STEP WAS EXECUTED - COND CODE 0000
IEF285I   SYS1.MACLIB                                KEPT          *-------24
IEF285I   VOL SER NOS= MVSRES.
IEF285I   SYS2.MACLIB                                KEPT          *--------0
IEF285I   VOL SER NOS= MVSRES.
IEF285I   SYS1.MACLIB                                KEPT          *--------0
IEF285I   VOL SER NOS= MVSRES.
IEF285I   SYS1.MACLIB                                KEPT          *--------0
IEF285I   VOL SER NOS= MVSRES.
IEF285I   SYS26305.T205049.RA000.HANOIASM.SYSUT1      DELETED       *------152
IEF285I   VOL SER NOS= WORK03.
IEF285I   SYS26305.T205049.RA000.HANOIASM.SYSUT2      DELETED       *-------29
IEF285I   VOL SER NOS= WORK02.
IEF285I   SYS26305.T205049.RA000.HANOIASM.SYSUT3      DELETED       *-------12
IEF285I   VOL SER NOS= WORK01.
IEF285I   JES2.JOB00955.SO0109                       SYSOUT
IEF285I   JES2.JOB00955.SO0110                       SYSOUT
IEF285I   SYS26305.T205049.RA000.HANOIASM.OBJSET      PASSED        *-------29
IEF285I   VOL SER NOS= WORK00.
IEF285I   JES2.JOB00955.SI0102                       SYSIN
IEF373I STEP /ASM     / START 26305.2050
IEF374I STEP /ASM     / STOP  26305.2050 CPU    0MIN 00.18SEC SRB    0MIN 00.02SEC VIRT   128K SYS   184K
********************************************************************************************************************************
*     2. Jobstep of job: HANOIASM    Stepname: ASM         Program name: IFOX00     Executed on 01.11.26 from 20.50.49 to 20.50.49 *
*         elapsed time  00:00:00,26                    CPU-Identifier:  TK4-           Page-in:      0                          *
*             CPU time  00:00:00,20                Virtual Storage used:    128K         Page-out:      0                          *
*           corr. CPU:  00:00:00,20   CPU time has been corrected by  1 / 1,0  multiplier                                        *
*                                                                                                                                *
*     I/O Operation                                                                                                              *
*     Number of records read via DD * or DD DATA:    371                                                                         *
*     148......24 148.......0 148.......0 148.......0 190.....152 180......29 170......12 DMY.......0 DMY.......0 140......29      *
*     DMY.......0                                                                                                                *
*                                                                                                                                *
*                                    Charge for step (w/o SYSOUT):          0,33                                                *
********************************************************************************************************************************
IEF236I ALLOC. FOR HANOIASM ASM HANOIASM
IEF237I 148  ALLOCATED TO SYSLIB
IEF237I 148  ALLOCATED TO
IEF237I 148  ALLOCATED TO
IEF237I 148  ALLOCATED TO
IEF237I 190  ALLOCATED TO SYSUT1
IEF237I 180  ALLOCATED TO SYSUT2
IEF237I 170  ALLOCATED TO SYSUT3
IEF237I JES2 ALLOCATED TO SYSPRINT
IEF237I JES2 ALLOCATED TO SYSPUNCH
IEF237I 140  ALLOCATED TO SYSGO
IEF237I JES2 ALLOCATED TO SYSIN
IEF142I HANOIASM ASM HANOIASM - STEP WAS EXECUTED - COND CODE 0000
IEF285I   SYS1.MACLIB                                KEPT          *-------43
IEF285I   VOL SER NOS= MVSRES.
IEF285I   SYS2.MACLIB                                KEPT          *--------0
IEF285I   VOL SER NOS= MVSRES.
```

```
IEF285I   SYS1.MACLIB                                  KEPT          *--------0
IEF285I   VOL SER NOS= MVSRES.
IEF285I   SYS1.MACLIB                                  KEPT          *--------0
IEF285I   VOL SER NOS= MVSRES.
IEF285I   SYS26305.T205049.RA000.HANOIASM.SYSUT1       DELETED       *------401
IEF285I   VOL SER NOS= WORK03.
IEF285I   SYS26305.T205049.RA000.HANOIASM.SYSUT2       DELETED       *-------40
IEF285I   VOL SER NOS= WORK02.
IEF285I   SYS26305.T205049.RA000.HANOIASM.SYSUT3       DELETED       *-------10
IEF285I   VOL SER NOS= WORK01.
IEF285I   JES2.JOB00955.SO0111                         SYSOUT
IEF285I   JES2.JOB00955.SO0112                         SYSOUT
IEF285I   SYS26305.T205049.RA000.HANOIASM.OBJSET       PASSED        *-------49
IEF285I   VOL SER NOS= WORK00.
IEF285I   JES2.JOB00955.SI0103                         SYSIN
IEF373I STEP /ASM     / START 26305.2050
IEF374I STEP /ASM     / STOP  26305.2050 CPU    0MIN 00.41SEC SRB    0MIN 00.05SEC VIRT   128K SYS   184K
*********************************************************************************************************
*     3. Jobstep of job: HANOIASM    Stepname: ASM         Program name: IFOX00     Executed on 01.11.26 from 20.50.49 to 20.50.50 *
*         elapsed time  00:00:00,56                    CPU-Identifier:  TK4-          Page-in:      0                          *
*             CPU time  00:00:00,46              Virtual Storage used:    128K        Page-out:     0                          *
*           corr. CPU:  00:00:00,46   CPU time has been corrected by  1 / 1,0  multiplier                                     *
*                                                                                                                             *
*     I/O Operation                                                                                                           *
*     Number of records read via DD * or DD DATA:    229                                                                      *
*     148......43 148.......0 148.......0 148.......0 190.....401 180......40 170......10 DMY.......0 DMY.......0 140......49 *
*     DMY.......0                                                                                                             *
*                                                                                                                             *
*                                       Charge for step (w/o SYSOUT):          0,76                                       *
*********************************************************************************************************
IEF236I ALLOC. FOR HANOIASM LKED HANOIASM
IEF237I 140  ALLOCATED TO SYSLIN
IEF237I JES2 ALLOCATED TO
IEF237I 180  ALLOCATED TO SYSLMOD
IEF237I 190  ALLOCATED TO SYSUT1
IEF237I JES2 ALLOCATED TO SYSPRINT
IEF237I JES2 ALLOCATED TO SYSIN
IEF142I HANOIASM LKED HANOIASM - STEP WAS EXECUTED - COND CODE 0000
IEF285I   SYS26305.T205049.RA000.HANOIASM.OBJSET       DELETED       *------101
IEF285I   VOL SER NOS= WORK00.
IEF285I   JES2.JOB00955.SI0104                         SYSIN
IEF285I   SYS26305.T205049.RA000.HANOIASM.GOSET        PASSED        *-------10
IEF285I   VOL SER NOS= WORK02.
IEF285I   SYS26305.T205049.RA000.HANOIASM.SYSUT1       DELETED       *--------0
IEF285I   VOL SER NOS= WORK03.
IEF285I   JES2.JOB00955.SO0113                         SYSOUT
IEF285I   JES2.JOB00955.SI0105                         SYSIN
IEF373I STEP /LKED    / START 26305.2050
IEF374I STEP /LKED    / STOP  26305.2050 CPU    0MIN 00.05SEC SRB    0MIN 00.01SEC VIRT   128K SYS   224K
*********************************************************************************************************
*     4. Jobstep of job: HANOIASM    Stepname: LKED        Program name: IEWL       Executed on 01.11.26 from 20.50.50 to 20.50.50 *
*         elapsed time  00:00:00,10                    CPU-Identifier:  TK4-          Page-in:      0                          *
*             CPU time  00:00:00,06              Virtual Storage used:    128K        Page-out:     0                          *
*           corr. CPU:  00:00:00,06   CPU time has been corrected by  1 / 1,0  multiplier                                     *
*                                                                                                                             *
*     I/O Operation                                                                                                           *
*     Number of records read via DD * or DD DATA:      2                                                                      *
*     140.....101 DMY.......0 180......10 190.......0 DMY.......0 DMY.......0                                                 *
*                                                                                                                             *
*                                       Charge for step (w/o SYSOUT):          0,10                                       *
*********************************************************************************************************
IEF236I ALLOC. FOR HANOIASM GO HANOIASM
IEF237I 180  ALLOCATED TO PGM=*.DD
IEF237I JES2 ALLOCATED TO HANOIIN
IEF237I JES2 ALLOCATED TO HANOIOUT
HANOIASM... Begin execution
```

```
HANOIASM... calling HANOIITR
HANOIITR... start HANOIITR
HANOIITR... end HANOIITR
HANOIASM... back from HANOIITR
HANOIASM... Normal end of program
IEF142I HANOIASM GO HANOIASM - STEP WAS EXECUTED - COND CODE 0000
IEF285I   SYS26305.T205049.RA000.HANOIASM.GOSET        KEPT          *--------0
IEF285I   VOL SER NOS= WORK02.
IEF285I   JES2.JOB00955.SI0106                         SYSIN
IEF285I   JES2.JOB00955.SO0114                         SYSOUT
IEF373I STEP /GO      / START 26305.2050
IEF374I STEP /GO      / STOP  26305.2050 CPU    0MIN 00.02SEC SRB    0MIN 00.00SEC VIRT    12K SYS   224K
*------------------------------------------------------------------------------------------------------------*
*     5. Jobstep of job: HANOIASM    Stepname: GO          Program name: PGM=*.DD   Executed on 01.11.26 from 20.50.50 to 20.50.50 *
*         elapsed time  00:00:00,03                    CPU-Identifier:  TK4-           Page-in:      0                        *
*             CPU time  00:00:00,02              Virtual Storage used:     12K         Page-out:     0                        *
*           corr. CPU:  00:00:00,02   CPU time has been corrected by  1 / 1,0  multiplier                                   *
*                                                                                                                            *
*     I/O Operation                                                                                                          *
*     Number of records read via DD * or DD DATA:      1                                                                     *
*     180.......0 DMY.......0 DMY.......0                                                                                    *
*                                                                                                                            *
*                                      Charge for step (w/o SYSOUT):          0,03                                       *
*------------------------------------------------------------------------------------------------------------*
IEF237I 180  ALLOCATED TO SYS00001
IEF285I   SYS26305.T205050.RA000.HANOIASM.R0000001     KEPT          *--------0
IEF285I   VOL SER NOS= WORK02.
IEF285I   SYS26305.T205049.RA000.HANOIASM.GOSET        DELETED
IEF285I   VOL SER NOS= WORK02.
IEF375I  JOB /HANOIASM/ START 26305.2050
IEF376I  JOB /HANOIASM/ STOP  26305.2050 CPU    0MIN 00.88SEC SRB    0MIN 00.10SEC
```

```
SYMBOL   TYPE  ID   ADDR  LENGTH LDID                                    ASM 0201 20.50 11/01/26

          PC  0001 000000 000000
HANOIREC  SD  0002 000000 0002A9
HANOIREC  ER  0003
```

```
LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                ASM 0201 20.50 11/01/26

                                    1 ********************************************************************** 00003101
                                    2 * ////////////////////////////////////////////////////////////////// 00003201
                                    3 * Title: HANOIREC - process steps recursively and  print slice movement 00003301
                                    4 *   lines                                                            00003421
                                    5 * Author: ManInBlueAT                                               00003501
                                    6 * Language: ASM                                                     00003601
                                    7 *                                                                   00003701
                                    8 * Description: Recursive function HANOIREC processes the following  00003821
                                    9 *   sequence of steps: Move the uppper n-1 slices from the source   00003921
                                   10 *   (from) to the auxiliary stack (aux), move the lowest slice (n)  00004021
                                   11 *   from the source to the destination stack (to) and move the upper 00004121
                                   12 *   n-1 slices from the aux to the destination stack. Apply this    00004221
                                   13 *   algorithm to all levels n recursively what results in 2**n-1 steps. 00004321
                                   14 *                                                                   00004401
                                   15 * Changes:                                                          00004501
                                   16 * - ManInBlueAT, 2026-10-09 New module                             00004601
                                   17 * ////////////////////////////////////////////////////////////////// 00004701
                                   18         PRINT GEN                                                  00004801
                                   19         REGISTER                                                   00004901
                        00000      20+R0      EQU   0                          , GENERAL PURPOSE REGISTER 0
                        00001      21+R1      EQU   1                          , GENERAL PURPOSE REGISTER 1
                        00002      22+R2      EQU   2                          , GENERAL PURPOSE REGISTER 2
                        00003      23+R3      EQU   3                          , GENERAL PURPOSE REGISTER 3
                        00004      24+R4      EQU   4                          , GENERAL PURPOSE REGISTER 4
                        00005      25+R5      EQU   5                          , GENERAL PURPOSE REGISTER 5
                        00006      26+R6      EQU   6                          , GENERAL PURPOSE REGISTER 6
                        00007      27+R7      EQU   7                          , GENERAL PURPOSE REGISTER 7
                        00008      28+R8      EQU   8                          , GENERAL PURPOSE REGISTER 8
                        00009      29+R9      EQU   9                          , GENERAL PURPOSE REGISTER 9
                        0000A      30+R10     EQU   10                         , GENERAL PURPOSE REGISTER 10
                        0000B      31+R11     EQU   11                         , GENERAL PURPOSE REGISTER 11
                        0000C      32+R12     EQU   12                         , GENERAL PURPOSE REGISTER 12
                        0000D      33+R13     EQU   13                         , GENERAL PURPOSE REGISTER 13
                        0000E      34+R14     EQU   14                         , GENERAL PURPOSE REGISTER 14
                        0000F      35+R15     EQU   15                         , GENERAL PURPOSE REGISTER 15
000000                             36 HANOIREC CSECT                                                     00005001
                                   37         SAVE  (14,12)                                              00005101
000000                             38+        DS    0H                                                   01650000
000000 90EC D00C      0000C        39+        STM   14,12,12(13)                  SAVE REGISTERS         02950000
000004 05C0                        40         BALR  R12,0                                                00005201
                        00006      41         USING *,R12                                                00005301
000006 1821                        42         LR    R2,R1              remember address of paramter list 00005421
                                   43 * allocate instance data                                           00005501
000008 4100 0058      00058        44         LA    R0,INSTL           length of area to be allocated    00005621
                                   45         GETMAIN R,LV=(0)         allocate area with length R0     00005701
                                   46+*      OS/VS2 RELEASE 4 VERSION -- 10/21/75                       00004804
00000C 4510 C00A      00010        47+        BAL   1,*+4                          INDICATE GETMAIN      00682402
000010 0A0A                        48+        SVC   10                             ISSUE GETMAIN SVC     00820002
000012 18B1                        49         LR    R11,R1             address of allocated storage in R1 00005821
                        00000      50         USING INSTANCE,R11       SAVEAREA is at the start of INSTANCE 00005921
000014 4130 B000      00000        51         LA    R3,SAVEAREA                                          00006001
000018 5030 D008      00008        52         ST    R3,8(,R13)                                           00006101
00001C 50D0 B004      00004        53         ST    R13,SAVEAREA+4                                       00006201
000020 18D3                        54         LR    R13,R3                                               00006301
                                   55 *--- handle parameters                                             00006421
```

```
  LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                ASM 0201 20.50 11/01/26

                                     56 *        WTO   'HANOIREC... start HANOIREC'                          00006501
000022 5830 2000      00000          57          L     R3,0(,R2)           reference address list            00006601
000026 5840 3000      00000          58          L     R4,0(,R3)           reference 1st address (HANOIOUT)  00006701
00002A 5040 B048      00048          59          ST    R4,HANOIOUT         store address in HANOIOUT         00006801
00002E 5840 3004      00004          60          L     R4,4(,R3)           reference 2nd address             00006901
000032 F811 B04C 4000 0004C 00000    61          ZAP   SLICE(2),0(2,R4)                                      00007001
000038 5840 3008      00008          62          L     R4,8(,R3)           reference 3nd address             00007101
00003C F800 B04E 4000 0004E 00000    63          ZAP   FROM(1),0(1,R4)                                      00007201
000042 5840 300C      0000C          64          L     R4,12(,R3)          reference 4th address             00007301
000046 F800 B04F 4000 0004F 00000    65          ZAP   TO(1),0(1,R4)                                        00007401
00004C 5840 3010      00010          66          L     R4,16(,R3)          reference 5th address             00007501
000050 F800 B050 4000 00050 00000    67          ZAP   AUX(1),0(1,R4)                                       00007601
000056 5840 3014      00014          68          L     R4,20(,R3)          reference 6th address             00007701
00005A F833 B051 4000 00051 00000    69          ZAP   STEPS(4),0(4,R4)                                     00007801
                                     70 * remember current slice value and current stack values as received  00007901
000060 F811 C1F6 B04C 001FC 0004C    71          ZAP   CSLICE,SLICE                                         00008001
000066 F800 C1F8 B04E 001FE 0004E    72          ZAP   CFROM,FROM                                           00008101
00006C F800 C1F9 B04F 001FF 0004F    73          ZAP   CTO,TO                                               00008201
000072 F800 C1FA B050 00200 00050    74          ZAP   CAUX,AUX                                             00008301
                                     75 * check if slice is not greater 1 - no recursive calls               00008401
000078 F910 C1F6 C2A2 001FC 002A8    76          CP    CSLICE,=P'1'                                         00008501
00007E 47D0 C154      0015A          77          BNP   DOSTEP             <=1 or overflow BC X'0C'          00008601
                                     78 * slice>1 --> call HANOIREC recursively two times with slice-1       00008701
                                     79 * put together argument values for 1st recursive call               00008801
                                     80 * HANOIREC(SLICE-1,FROM,AUX,TO,STEPS)                               00008901
                                     81 * set addresses of parameters                                       00009001
000082 5840 B048      00048          82          L     R4,HANOIOUT                                          00009101
000086 5040 C1DE      001E4          83          ST    R4,AHANOUT                                           00009201
00008A 4140 B04C      0004C          84          LA    R4,SLICE                                             00009301
00008E 5040 C1E2      001E8          85          ST    R4,ASLICE                                            00009401
000092 4140 B04E      0004E          86          LA    R4,FROM                                              00009501
000096 5040 C1E6      001EC          87          ST    R4,AFROM                                             00009601
00009A 4140 B04F      0004F          88          LA    R4,TO                                                00009701
00009E 5040 C1EA      001F0          89          ST    R4,ATO                                               00009801
0000A2 4140 B050      00050          90          LA    R4,AUX                                               00009901
0000A6 5040 C1EE      001F4          91          ST    R4,AAUX                                              00010001
0000AA 4140 B051      00051          92          LA    R4,STEPS                                             00010101
0000AE 5040 C1F2      001F8          93          ST    R4,ASTEPS                                            00010201
                                     94 * set slice and stack values in instance area                       00010301
0000B2 FB10 B04C C2A2 0004C 002A8    95          SP    SLICE,=P'1'         decrement slice                  00010401
                                     96 *        ZAP   FROM,CFROM          no change                        00010501
0000B8 F800 B04F C1FA 0004F 00200    97          ZAP   TO,CAUX                                              00010601
0000BE F800 B050 C1F9 00050 001FF    98          ZAP   AUX,CTO                                              00010701
                                     99 * call HANOIREC recursively - 1st call                              00010801
                                    100          CALL  HANOIREC,(ALIST)                                     00010901
0000C4                              101+         CNOP  0,4                                                  00750000
0000C4 47F0 C0C6      000CC         102+         B     *+8                 BRANCH AROUND VCON               00800000
0000C8 00000000                     103+IHB0004B DC    V(HANOIREC)         ENTRY POINT ADDRESS              00850000
0000CC                              104+         CNOP  0,4                                                  02800000
0000CC 4510 C0CE      000D4         105+         BAL   1,IHB0005A          LOAD LIST ADDR IN REG1           02850000
                            000D0   106+IHB0005  EQU   *                                                    02900000
0000D0 00                           107+         DC    B'00000000'         SET VL SWITCH BIT                03700000
0000D1 0001E4                       108+         DC    AL3(ALIST)          PROB. PROG. PARAMETER            03750000
                            000D4   109+IHB0005A EQU   *                                                    04100000
0000D4 58F0 C0C2      000C8         110+         L     15,IHB0004B         LOAD 15 WITH ENTRY ADR           01100000
```

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  LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                                ASM 0201 20.50 11/01/26

0000D8 05EF                       111+         BALR  14,15                                BRANCH TO ENTRY POINT  01150000
                                  112 * recover received slice and stack values from instance area                  00011001
                                  113 * as Cxxxxx values are no longer valid                                        00011101
0000DA F811 C1F6 B04C 001FC 0004C 114          ZAP   CSLICE,SLICE                                                 00011201
0000E0 FA10 C1F6 C2A2 001FC 002A8 115          AP    CSLICE,=P'1'        increment slice (current level)          00011321
0000E6 F800 C1F8 B04E 001FE 0004E 116          ZAP   CFROM,FROM                                                   00011401
0000EC F800 C1FA B04F 00200 0004F 117          ZAP   CAUX,TO                                                      00011501
0000F2 F800 C1F9 B050 001FF 00050 118          ZAP   CTO,AUX                                                      00011601
                                  119 * do the step                                                              00011701
0000F8 47F0 C154      0015A       120          B     DOSTEP                                                       00011801
                          000FC   121 DO2ND    EQU   *                                                            00011901
                                  122 * set stack valaues for recursive call 2                                   00012001
                                  123 * HANOIREC(SLICE-1,AUX,TO,FROM,STEPS)                                      00012101
                                  124 * set addresses of parameters                                              00012201
0000FC 5840 B048      00048       125          L     R4,HANOIOUT                                                  00012301
000100 5040 C1DE      001E4       126          ST    R4,AHANOUT                                                   00012401
000104 4140 B04C      0004C       127          LA    R4,SLICE                                                     00012521
000108 5040 C1E2      001E8       128          ST    R4,ASLICE                                                    00012601
00010C 4140 B04E      0004E       129          LA    R4,FROM                                                      00012701
000110 5040 C1E6      001EC       130          ST    R4,AFROM                                                     00012801
000114 4140 B04F      0004F       131          LA    R4,TO                                                        00012901
000118 5040 C1EA      001F0       132          ST    R4,ATO                                                       00013001
00011C 4140 B050      00050       133          LA    R4,AUX                                                       00013101
000120 5040 C1EE      001F4       134          ST    R4,AAUX                                                      00013201
000124 4140 B051      00051       135          LA    R4,STEPS                                                     00013301
000128 5040 C1F2      001F8       136          ST    R4,ASTEPS                                                    00013401
                                  137 * set slice and stack values in instance area                              00013501
                                  138 *        ZAP   SLICE,CSLICE        no change                            00013601
                                  139 *        SP    SLICE,=P'1'         no change, already decremented       00013721
00012C F800 B04E C1FA 0004E 00200 140          ZAP   FROM,CAUX                                                    00013801
000132 F800 B04F C1F9 0004F 001FF 141          ZAP   TO,CTO                                                       00013901
000138 F800 B050 C1F8 00050 001FE 142          ZAP   AUX,CFROM                                                    00014001
                                  143 * call HANOILN recursively - 2nd call                                     00014101
                                  144          CALL  HANOIREC,(ALIST)                                           00014201
00013E 0700                       145+         CNOP  0,4                                                        00750000
000140 47F0 C142      00148       146+         B     *+8                             BRANCH AROUND VCON       00800000
000144 00000000                   147+IHB0006B DC    V(HANOIREC)                     ENTRY POINT ADDRESS      00850000
000148                            148+         CNOP  0,4                                                        02800000
000148 4510 C14A      00150       149+         BAL   1,IHB0007A                      LOAD LIST ADDR IN REG1   02850000
                          0014C   150+IHB0007  EQU   *                                                        02900000
00014C 00                         151+         DC    B'00000000'                     SET VL SWITCH BIT        03700000
00014D 0001E4                     152+         DC    AL3(ALIST)                      PROB. PROG. PARAMETER    03750000
                          00150   153+IHB0007A EQU   *                                                        04100000
000150 58F0 C13E      00144       154+         L     15,IHB0006B                     LOAD 15 WITH ENTRY ADR   01100000
000154 05EF                       155+         BALR  14,15                           BRANCH TO ENTRY POINT    01150000
                                  156 * go to end of segment                                                    00014301
000156 47F0 C1BC      001C2       157          B     PROCEND             end of segment                       00014401
                                  158 *---                                                                      00014501
                                  159 * Perform the step of moving a slice. The counter steps is increased,     00014621
                                  160 * current slice cslice is moved from the current source stack cfrom       00014701
                                  161 * to the current destination stack cto                                    00014801
                          0015A   162 DOSTEP   EQU   *                                                        00014901
00015A FA30 B051 C2A2 00051 002A8 163          AP    STEPS,=P'1'         increment steps                      00015021
                                  164 * prepare step line record                                               00015101
000160 D207 C28C C294 00292 0029A 165          MVC   TRGTSTPS,TMPSTPS                                           00015201
```

```
  LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                  ASM 0201 20.50 11/01/26

000166 DE07 C28C B051 00292 00051   166          ED    TRGTSTPS,STEPS                                        00015301
00016C D206 C201 C28D 00207 00293   167          MVC   OSTEP,TRGTSTPS+1                                      00015401
000172 D203 C280 C284 00286 0028A   168          MVC   TRGTSLC,TMPSLC                                        00015501
000178 DE03 C280 C1F6 00286 001FC   169          ED    TRGTSLC,CSLICE                                        00015601
00017E D201 C217 C282 0021D 00288   170          MVC   OSLICE,TRGTSLC+2                                      00015701
000184 D201 C288 C28A 0028E 00290   171          MVC   TRGTSTK,TMPSTK                                        00015801
00018A DE01 C288 C1F8 0028E 001FE   172          ED    TRGTSTK,CFROM                                         00015901
000190 D200 C225 C289 0022B 0028F   173          MVC   OFROM,TRGTSTK+1                                       00016001
000196 D201 C288 C28A 0028E 00290   174          MVC   TRGTSTK,TMPSTK                                        00016101
00019C DE01 C288 C1F9 0028E 001FF   175          ED    TRGTSTK,CTO                                           00016201
0001A2 D200 C22A C289 00230 0028F   176          MVC   OTO,TRGTSTK+1                                         00016301
0001A8 5840 B048      00048         177          L     R4,HANOIOUT         load address of HANOIOUT DCB      00016401
                                    178          PUT   (R4),OREC           write step line                   00016501
0001AC 1814                         179+         LR    1,R4                          LOAD PARAMETER REG 1    02050002
0001AE 4100 C1FB      00201         180+         LA    0,OREC                        LOAD PARAMETER REG 0    02500002
0001B2 58F0 1030      00030         181+         L     15,48(0,1)              LOAD PUT ROUTINE ADDR         00550000
0001B6 05EF                         182+         BALR  14,15                   LINK TO PUT ROUTINE           00600000
                                    183 * do 2nd recursive call after step if current slice number>1        00016601
0001B8 F910 C1F6 C2A2 001FC 002A8   184          CP    CSLICE,=P'1'                                          00016701
0001BE 4720 C0F6      000FC         185          BP    DO2ND               >1 BC X'02'                       00016801
                                    186 * end of segment processing                                         00016901
                           001C2    187 PROCEND  EQU   *                                                    00017021
                                    188 * return to caller actions                                          00017101
0001C2 58D0 B004      00004         189          L     R13,SAVEAREA+4                                        00017201
                                    190 * return current number of steps in caller INSTANCE structure       00017301
0001C6 F833 D051 B051 00051 00051   191          ZAP   STEPSPOS(4,R13),STEPS(4)                              00017403
                                    192 * free current INSTANCE area                                        00017501
0001CC 4110 B000      00000         193          LA    R1,INSTANCE                                           00017601
0001D0 4100 0058      00058         194          LA    R0,INSTL            length of allocated area          00017705
                                    195          FREEMAIN R,LV=(0),A=(1)   free area R1 with length R0       00017801
                                    196+*      OS/VS2 RELEASE 3 VERSION  -- 10/25/74                        00001603
0001D4 4110 1000      00000         197+         LA    1,0(0,1)                      CLEAR HI ORDER BYTE     00150802
0001D8 0A0A                         198+         SVC   10                            ISSUE FREEMAIN SVC      00311202
                                    199 *        WTO   'HANOIREC... end HANOIREC'                            00017905
                                    200          RETURN (14,12),RC=0                                         00018001
0001DA 98EC D00C      0000C         201+         LM    14,12,12(13)                  RESTORE THE REGISTERS   00650000
0001DE 41F0 0000      00000         202+         LA    15,0(0,0)                     LOAD RETURN CODE        01750000
0001E2 07FE                         203+         BR    14                            RETURN                  02000000
                                    204 *--- data definitions                                               00018101
                                    205 * list of parameters (for recursive call of HANOIREC)               00018201
0001E4                              206 ALIST    DS    0A                                                    00018301
0001E4                              207 AHANOUT  DS    A                                                     00018401
0001E8                              208 ASLICE   DS    A                                                     00018501
0001EC                              209 AFROM    DS    A                                                     00018601
0001F0                              210 ATO      DS    A                                                     00018701
0001F4                              211 AAUX     DS    A                                                     00018801
0001F8                              212 ASTEPS   DS    A                                                     00018901
                                    213 * remember current slice and stack values as received               00019001
0001FC                              214 CSLICE   DS    PL2                 slice value                       00019101
0001FE                              215 CFROM    DS    PL1                 from value                        00019201
0001FF                              216 CTO      DS    PL1                 to value                          00019301
000200                              217 CAUX     DS    PL1                 auxiliary value                   00019401
                                    218 * savearea and parameters (and local data of the instance, if any)  00019501
                                    219 * to be allocated before recursive calls                            00019621
000000                              220 INSTANCE DSECT                                                      00019701
```

```
  LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                  ASM 0201 20.50 11/01/26

000000                             221 SAVEAREA DS    18F                                                  00019801
000048                             222 HANOIOUT DS    A(4)               address of output file DCB        00019901
00004C                             223 SLICE    DS    PL2                                                  00020001
00004E                             224 FROM     DS    PL1                                                  00020101
00004F                             225 TO       DS    PL1                                                  00020201
000050                             226 AUX      DS    PL1                                                  00020301
000051                             227 STEPS    DS    PL4                                                  00020401
000055 404040                      228          DC    CL3'   '           alignment                        00020521
                           00058   229 INSTL    EQU   *-INSTANCE                                           00020601
                           00051   230 STEPSPOS EQU   STEPS-INSTANCE                                       00020701
                                   231 * end DSECT, continue CSECT                                         00020801
000201                             232 HANOIREC CSECT                                                      00020901
                                   233 * output line definition for step                                   00021001
000201                             234 OREC     DS    0CL133                                               00021101
000201 40                          235          DC    CL1' '              ASO character                    00021201
000202 E2A3859740                  236          DC    CL5'Step '                                           00021301
000207                             237 OSTEP    DS    CL7                 current step number              00021401
00020E 7A40D496A5899587            238          DC    CL15': Moving slice '                                00021501
00021D                             239 OSLICE   DS    CL2                 current slice                    00021601
00021F 408699969440A2A3            240          DC    CL12' from stack '                                   00021701
00022B                             241 OFROM    DS    CL1                 from stack                       00021801
00022C 40A39640                    242          DC    CL4' to '                                            00021901
000230                             243 OTO      DS    CL1                 to stack                         00022001
000231 4040404040404040            244          DC    CL85' '                                              00022101
                                   245 * templates and target fields for print preparation                  00022201
000286                             246 TRGTSLC  DS    CL4                                                  00022301
00028A 40202120                    247 TMPSLC   DC    X'40202120'                                          00022419
00028E                             248 TRGTSTK  DS    CL2                                                  00022501
000290 4020                        249 TMPSTK   DC    X'4020'                                              00022619
000292                             250 TRGTSTPS DS    CL8                                                  00022701
00029A 4020202020202120            251 TMPSTPS  DC    X'4020202020202120'                                  00022819
                                   252 *--- literals, if any, will go here                                  00022901
0002A8                             253          LTORG *                                                   00023001
0002A8 1C                          254                =P'1'
                                   255 *--- end of section                                                 00023101
000000                             256          END   HANOIREC                                             00023201
```

POS.ID	REL.ID	FLAGS	ADDRESS
0002	0002	08	0000D1
0002	0002	08	00014D
0002	0003	1C	0000C8
0002	0003	1C	000144

SYMBOL	LEN	VALUE	DEFN	REFERENCES
AAUX	00004	000001F4	00211	00091 00134
AFROM	00004	000001EC	00209	00087 00130
AHANOUT	00004	000001E4	00207	00083 00126
ALIST	00004	000001E4	00206	00108 00152
ASLICE	00004	000001E8	00208	00085 00128
ASTEPS	00004	000001F8	00212	00093 00136
ATO	00004	000001F0	00210	00089 00132
AUX	00001	00000050	00226	00067 00074 00090 00098 00118 00133 00142
CAUX	00001	00000200	00217	00074 00097 00117 00140
CFROM	00001	000001FE	00215	00072 00116 00142 00172
CSLICE	00002	000001FC	00214	00071 00076 00114 00115 00169 00184
CTO	00001	000001FF	00216	00073 00098 00118 00141 00175
DOSTEP	00001	0000015A	00162	00077 00120
DO2ND	00001	000000FC	00121	00185
FROM	00001	0000004E	00224	00063 00072 00086 00116 00129 00140
HANOIOUT	00004	00000048	00222	00059 00082 00125 00177
HANOIREC	00001	00000000	00036	00232 00256
IHB0004B	00004	000000C8	00103	00110
IHB0005A	00001	000000D4	00109	00105
IHB0006B	00004	00000144	00147	00154
IHB0007A	00001	00000150	00153	00149
INSTANCE	00001	00000000	00220	00050 00193 00229 00230
INSTL	00001	00000058	00229	00044 00194
OFROM	00001	0000022B	00241	00173
OREC	00133	00000201	00234	00180
OSLICE	00002	0000021D	00239	00170
OSTEP	00007	00000207	00237	00167
OTO	00001	00000230	00243	00176
PROCEND	00001	000001C2	00187	00157
R0	00001	00000000	00020	00044 00194
R1	00001	00000001	00021	00042 00049 00193
R11	00001	0000000B	00031	00049 00050
R12	00001	0000000C	00032	00040 00041
R13	00001	0000000D	00033	00052 00053 00054 00189 00191
R2	00001	00000002	00022	00042 00057
R3	00001	00000003	00023	00051 00052 00054 00057 00058 00060 00062 00064 00066 00068
R4	00001	00000004	00024	00058 00059 00060 00061 00062 00063 00064 00065 00066 00067 00068 00069 00082 00083 00084
				00085 00086 00087 00088 00089 00090 00091 00092 00093 00125 00126 00127 00128 00129 00130
				00131 00132 00133 00134 00135 00136 00177 00179
SAVEAREA	00004	00000000	00221	00051 00053 00189
SLICE	00002	0000004C	00223	00061 00071 00084 00095 00114 00127
STEPS	00004	00000051	00227	00069 00092 00135 00163 00166 00191 00230
STEPSPOS	00004	00000051	00230	00191
TMPSLC	00004	0000028A	00247	00168
TMPSTK	00002	00000290	00249	00171 00174
TMPSTPS	00008	0000029A	00251	00165
TO	00001	0000004F	00225	00065 00073 00088 00097 00117 00131 00141
TRGTSLC	00004	00000286	00246	00168 00169 00170
TRGTSTK	00002	0000028E	00248	00171 00172 00173 00174 00175 00176
TRGTSTPS	00008	00000292	00250	00165 00166 00167

SYMBOL	LEN	VALUE	DEFN	REFERENCES
=P'1'	00001	000002A8	00254	00076 00095 00115 00163 00184

```
NO STATEMENTS FLAGGED IN THIS ASSEMBLY
HIGHEST SEVERITY WAS    0
OPTIONS FOR THIS ASSEMBLY
  ALIGN, ALOGIC, BUFSIZE(STD), NODECK, ESD, FLAG(0), LINECOUNT(55), LIST, NOMCALL, YFLAG, WORKSIZE(2097152)
  NOMLOGIC, NONUMBER, OBJECT, NORENT, RLD, NOSTMT, NOLIBMAC, NOTERMINAL, NOTEST, XREF(SHORT)
  SYSPARM()
WORK FILE BUFFER SIZE/NUMBER = 8702/ 1
TOTAL RECORDS READ FROM SYSTEM INPUT        202
TOTAL RECORDS READ FROM SYSTEM LIBRARY     1827
TOTAL RECORDS PUNCHED                        22
TOTAL RECORDS PRINTED                       346
```

SYMBOL TYPE ID ADDR LENGTH LDID ASM 0201 20.50 11/01/26

```
          PC  0001 000000 000000
HANOIITR  SD  0002 000000 00050C
```

```
LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                ASM 0201 20.50 11/01/26

                                    1 ********************************************************************** 00024001
                                    2 * ////////////////////////////////////////////////////////////////// 00024101
                                    3 * Title: HANOIITR - process steps iteratively and  print slice movement 00024201
                                    4 *        lines                                                       00024301
                                    5 * Author: ManInBlueAT                                               00024401
                                    6 * Language: ASM                                                     00024501
                                    7 *                                                                   00024601
                                    8 * Description: Iterative function HANOIITR to process the 2**n-1    00024721
                                    9 *   steps. There are sequences of three categories of movement starting 00024821
                                   10 *   with from stack to to stack or auxiliary stack depending on an odd  00024921
                                   11 *   or even number of slices. The direction of any movement is          00025021
                                   12 *   dependent on the sizes of the top slices of the involved from and   00025121
                                   13 *   to stacks.                                                          00025221
                                   14 *                                                                       00025321
                                   15 * Changes:                                                            00025401
                                   16 * - ManInBlueAT, 2026-10-09 New module                                00025501
                                   17 * ////////////////////////////////////////////////////////////////// 00025601
                                   18          MACRO                                                        00025719
                                   19 &NAME    SWAPF &FIXED41,&FIXED42   swap binary fixed 4 bytes values 1,2 00025819
                                   20          XC    &FIXED41,&FIXED42                                      00025919
                                   21          XC    &FIXED42,&FIXED41                                      00026019
                                   22          XC    &FIXED41,&FIXED42                                      00026119
                                   23          MEND                                                         00026219
                                   24 *-------------------------------------------------------------------- 00026319
                                   25          PRINT GEN                                                    00026401
                                   26          REGISTER                                                     00026501
                            00000  27+R0       EQU   0                          , GENERAL PURPOSE REGISTER 0
                            00001  28+R1       EQU   1                          , GENERAL PURPOSE REGISTER 1
                            00002  29+R2       EQU   2                          , GENERAL PURPOSE REGISTER 2
                            00003  30+R3       EQU   3                          , GENERAL PURPOSE REGISTER 3
                            00004  31+R4       EQU   4                          , GENERAL PURPOSE REGISTER 4
                            00005  32+R5       EQU   5                          , GENERAL PURPOSE REGISTER 5
                            00006  33+R6       EQU   6                          , GENERAL PURPOSE REGISTER 6
                            00007  34+R7       EQU   7                          , GENERAL PURPOSE REGISTER 7
                            00008  35+R8       EQU   8                          , GENERAL PURPOSE REGISTER 8
                            00009  36+R9       EQU   9                          , GENERAL PURPOSE REGISTER 9
                            0000A  37+R10      EQU   10                         , GENERAL PURPOSE REGISTER 10
                            0000B  38+R11      EQU   11                         , GENERAL PURPOSE REGISTER 11
                            0000C  39+R12      EQU   12                         , GENERAL PURPOSE REGISTER 12
                            0000D  40+R13      EQU   13                         , GENERAL PURPOSE REGISTER 13
                            0000E  41+R14      EQU   14                         , GENERAL PURPOSE REGISTER 14
                            0000F  42+R15      EQU   15                         , GENERAL PURPOSE REGISTER 15
000000                             43 HANOIITR CSECT                                                        00026601
                                   44          SAVE  (14,12)                                                00026701
000000                             45+         DS    0H                                                     01650000
000000 90EC D00C      0000C        46+         STM   14,12,12(13)                    SAVE REGISTERS         02950000
000004 05C0                        47          BALR  R12,0                                                  00026801
                            00006  48          USING *,R12                                                  00026901
000006 1821                        49          LR    R2,R1                remember address to paramter list 00027001
000008 4130 C3E2      003E8        50          LA    R3,SAVEAREA                                            00027101
00000C 5030 D008      00008        51          ST    R3,8(,R13)                                             00027201
000010 50D0 C3E6      003EC        52          ST    R13,SAVEAREA+4                                         00027301
000014 18D3                        53          LR    R13,R3                                                 00027401
                                   54 *--- handle parameters                                              00027501
                                   55          WTO   'HANOIITR... start HANOIITR'                           00027602
```

```
  LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                    ASM 0201 20.50 11/01/26

000016 0700                          56+         CNOP  0,4                                                        09800002
000018 4510 C034      0003A          57+         BAL   1,IHB0003A                    BRANCH AROUND MESSAGE  09850002
00001C 001E                          58+         DC    AL2(30)             TEXT LENGTH                        13200002
00001E 0000                          59+         DC    B'0000000000000000' MCS FLAGS                          13250002
000020 C8C1D5D6C9C9E3D9              60+         DC    C'HANOIITR... start HANOIITR'                          13350002
00003A                               61+IHB0003A DS    0H                                                     15850002
00003A 0A23                          62+         SVC   35                                                     15950002
00003C 5830 2000      00000          63          L     R3,0(,R2)           reference address list             00027701
000040 5840 3000      00000          64          L     R4,0(,R3)           reference 1st address (HANOIOUT)   00027801
000044 5040 C42A      00430          65          ST    R4,HANOIOUT         store address in HANOIOUT          00027901
000048 5840 3004      00004          66          L     R4,4(,R3)           reference 2nd address              00028001
00004C F811 C42E 4000 00434 00000    67          ZAP   SLICES(2),0(2,R4)                                      00028104
000052 5840 3008      00008          68          L     R4,8(,R3)           reference 3nd address              00028201
000056 F800 C430 4000 00436 00000    69          ZAP   FROM(1),0(1,R4)                                        00028301
00005C 5840 300C      0000C          70          L     R4,12(,R3)          reference 4th address              00028401
000060 F800 C431 4000 00437 00000    71          ZAP   TO(1),0(1,R4)                                          00028501
000066 5840 3010      00010          72          L     R4,16(,R3)          reference 5th address              00028601
00006A F800 C432 4000 00438 00000    73          ZAP   AUX(1),0(1,R4)      (initialized by caller)            00028705
000070 5840 3014      00014          74          L     R4,20(,R3)          reference 6th address              00028801
000074 F833 C433 4000 00439 00000    75          ZAP   STEPS(4),0(4,R4)    (initialized by caller to 0)       00028919
00007A 5040 C43A      00440          76          ST    R4,ASTEPS           remember address of i/o parm steps 00029003
                                     77 * convert slices parm to binary - value aligned, convert, store binary  00029121
00007E F871 C3DA C42E 003E0 00434    78          ZAP   TPALIGN,SLICES                                         00029210
000084 4F30 C3DA      003E0          79          CVB   R3,TPALIGN                                             00029310
000088 5030 C43E      00444          80          ST    R3,BSLICES                                             00029410
                                     81 *--- allocate dynamic stack data                                       00029521
                                     82 * get length area to be allocated: STKDYNL*3*(SLICES+1)               00029621
00008C 5830 C43E      00444          83            L     R3,BSLICES          no of slices                     00029711
000090 5A30 C4F2      004F8          84            A     R3,=F'1'            no of slices+1                   00029811
000094 4140 0004      00004          85            LA    R4,STKDYNL          dynamic stack element (slice)    00029917
000098 1C24                          86            MR    R2,R4               dynamic slices storage per stack 00030017
00009A 5030 C3B6      003BC          87            ST    R3,STKSLCS1         store binary storage length slices+1 00030111
00009E 5C20 C4F6      004FC          88            M     R2,=F'3'            storage for dynamic part 3 stacks 00030211
0000A2 5030 C3BA      003C0          89            ST    R3,STKALLOC         remember length of allocated storage 00030311
                                     90            GETMAIN R,LV=(3)          allocate area with length R3     00030411
                                     91+*      OS/VS2 RELEASE 4 VERSION -- 10/21/75                         00004804
0000A6 1803                          92+           LR    0,3                                 LOAD LENGTH  00646000
0000A8 4510 C0A6      000AC          93+           BAL   1,*+4                               INDICATE GETMAIN 00638002
0000AC 0A0A                          94+           SVC   10                                  ISSUE GETMAIN SVC 00820002
0000AE 18B1                          95            LR    R11,R1              R11 points to the allocated area 00030511
                           00000     96            USING STKDYN,R11          using R11 area for dynamic stk data 00030611
                                     97 *--- initialize STK (stack) structure                                  00030720
                                     98 * Internally we always move slices from from stack 1 to to stack 2     00030807
                                     99 * with auxiliary stack aux 3 if slices is odd and from from stack 1    00030907
                                    100 * to to stack 3 with aux stack 2 if slices is even. We assign the      00031007
                                    101 * stack names given as parameters as display names accordingly.       00031121
                                    102 * As the index orgin in AL is 0 the array indices of the STK data     00031207
                                    103 * is 0 to 2 (so we use stack 0,1,2)                                   00031320
0000B0 F800 C3A6 C430 003AC 00436   104           ZAP   STKDSPNO,FROM       display from stack to STK 0      00031407
0000B6 5850 C43E      00444         105           L     R5,BSLICES                                           00031517
0000BA 5450 C4F2      004F8         106           N     R5,=F'1'            check if no of slices is odd      00031617
0000BE 4780 C0CC      000D2         107           BZ    CONTEVEN            go to even if bit not set on pos 1 00031707
0000C2 F800 C3A7 C431 003AD 00437   108           ZAP   STKDSPNO+STKDSPNL,TO    display to stack to STK 1     00031807
0000C8 F800 C3A8 C432 003AE 00438   109           ZAP   STKDSPNO+2*STKDSPNL,AUX display aux stack to STK 2   00031907
0000CE 47F0 C0D8      000DE         110           B     CONTINIT                                             00032007
```

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  LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                ASM 0201 20.50 11/01/26

                           000D2   111 CONTEVEN EQU   *                                                        00032107
0000D2 F800 C3A7 C432 003AD 00438   112          ZAP   STKDSPNO+STKDSPNL,AUX  display aux stack to STK 1    00032207
0000D8 F800 C3A8 C431 003AE 00437   113          ZAP   STKDSPNO+2*STKDSPNL,TO display to stack to STK 2     00032307
                           000DE   114 CONTINIT EQU   *                                                        00032407
                                   115 * For every stack, maintain the count of slices and track the slices    00032507
                                   116 * by their position numbers (sizes) on the stack. Initially, all slices 00032621
                                   117 * are on stack 0.                                                       00032721
0000DE D203 C3AA C43E 003B0 00444   118          MVC   STKCOUNT(4),BSLICES STK 0 count starts with all slices 00032819
0000E4 D203 C3AE C4FA 003B4 00500   119          MVC   (STKCOUNT+STKCOUNL)(4),=X'00000000' STK 1 count set to 0 00032915
0000EA D203 C3B2 C4FA 003B8 00500   120          MVC   (STKCOUNT+2*STKCOUNL)(4),=X'00000000' STK 2 count set 0  00033015
                                   121 * The slices are maintained in STKSLICE array per stack. The indices    00033108
                                   122 * used are 1 to slices. The slice with index 0 (no real slice) is set   00033221
                                   123 * to a value (99) higher than any specified input value for slices.     00033321
                                   124 * So, every slice number (representing the "size" of the slice) on a    00033421
                                   125 * stack will always be smaller. In any situation the smallest slice     00033521
                                   126 * (least size number) has the top position (highest slice index on the  00033621
                                   127 * stack, >=1)                                                          00033721
0000F0 4120 0063      00063        128          LA    R2,99               slice imaginary high (size) value    00033821
0000F4 5020 B000      00000        129          ST    R2,STKSLICE         set value to STKSLICE element 0      00033911
0000F8 5830 C3B6      003BC        130          L     R3,STKSLCS1         dynamic storage slices per stack     00034012
0000FC 5023 B000      00000        131          ST    R2,STKSLICE(R3)     set value to stack 1 element 0       00034112
000100 5A30 C3B6      003BC        132          A     R3,STKSLCS1         dynamic storage slices per stack     00034221
000104 5023 B000      00000        133          ST    R2,STKSLICE(R3)     set value to stack 2 element 0       00034312
                                   134 * initialize slices 1 to BSLICES for every stack                       00034411
000108 5820 C43E      00444        135          L     R2,BSLICES          initial i                            00034512
00010C 5830 C4F2      004F8        136          L     R3,=F'1'            initial value for slices-i+1         00034612
000110 1852                        137          LR    R5,R2               i                                    00034712
000112 4160 0004      00004        138          LA    R6,STKDYNL          dynamic stack element (slice)        00034817
000116 1C46                        139          MR    R4,R6               STKDYNL*i initial displacement       00034917
000118 4140 0000      00000        140          LA    R4,0                R4 (was and) is 0                    00035019
00011C 5920 C4FE      00504        141 SLCINIT  C     R2,=X'00000001'                                          00035120
000120 4740 C14A      00150        142          BL    ENDINIT                                                  00035218
                                   143 * slice(0,i)=slices-i+1 --> the nearer the ground the larger the slice  00035312
000124 5035 B000      00000        144          ST    R3,STKSLICE(R5)     slices-i+1 to stk 0 offset i         00035412
                                   145 * initialize element slices-i+1 for stack 1,2 with 0 as these stacks    00035512
                                   146 * are empty at begin                                                   00035612
000128 4160 B000      00000        147          LA    R6,STKSLICE         basis for elements stack 1 and 2     00035712
00012C 1A65                        148          AR    R6,R5               + displacement current slice element 00035821
00012E 5A60 C3B6      003BC        149          A     R6,STKSLCS1         move to position stack 1             00035912
000132 5046 0000      00000        150          ST    R4,0(R6)            initialize stk 1 element with 0      00036012
000136 5A60 C3B6      003BC        151          A     R6,STKSLCS1         move to position stack 2             00036112
00013A 5046 0000      00000        152          ST    R4,0(R6)            initialize with 0                    00036212
                                   153 * prepare next iteration                                               00036312
00013E 5B20 C4F2      004F8        154          S     R2,=F'1'            i=i-1                                00036412
000142 5A30 C4F2      004F8        155          A     R3,=F'1'            next larger slice                    00036512
000146 4160 0004      00004        156          LA    R6,STKDYNL                                               00036617
00014A 1B56                        157          SR    R5,R6               stack 0, next lower position         00036717
00014C 47F0 C116      0011C        158          B     SLCINIT                                                  00036812
                           00150   159 ENDINIT  EQU   *                                                        00036918
                                   160 * perform 2**slices-1 steps to move the slices                         00037013
000150 4120 0001      00001        161          LA    R2,1                                                     00037113
000154 5830 C43E      00444        162          L     R3,BSLICES                                               00037213
000158 8B20 3000      00000        163          SLA   R2,0(R3)            R2 is now 2**slices                  00037318
00015C 4130 0001      00001        164          LA    R3,1                current step initialized with 1      00037421
000160 5030 C3BE      003C4        165          ST    R3,R3HELP           R3HELP shadows R3                    00037521
```

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  LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                ASM 0201 20.50 11/01/26

                                    166 * --- perform the steps loop                                        00037621
                                    167 * Do series of three different moves until done: 0 to 1 or 0 to 2  00037713
                                    168 * or 1 to 2 or vice versa (category 1, 2, 0)                       00037813
                                    169 * category = residue class 0,1,2 of step modulo 3 starting with 1  00037921
000164 4140 0001      00001         170          LA    R4,1                start with move category R4 1    00038014
000168 5920 C3BE      003C4         171 STPLOOP  C     R2,R3HELP                                            00038114
00016C 47D0 C352      00358         172          BNH   STPLPEND            R3HELP is 2**slices --> stop     00038219
000170 5940 C4FE      00504         173 MOVEC1   C     R4,=X'00000001'     category 1 move: 0-1 or vice versa 00038320
000174 4770 C186      0018C         174          BNE   MOVEC2                                               00038413
000178 D203 C442 C4FA 00448 00500   175          MVC   BFROM(4),=X'00000000' from stack 0                   00038518
00017E D203 C446 C4FE 0044C 00504   176          MVC   BTO(4),=X'00000001' to stack 1                       00038618
000184 4140 0002      00002         177          LA    R4,2                next move is category R4 2       00038714
000188 47F0 C1B2      001B8         178          B     STPLPCNT                                             00038814
00018C 5940 C502      00508         179 MOVEC2   C     R4,=X'00000002'     category 2 move: 0-2 or vice versa 00038920
000190 4770 C1A2      001A8         180          BNE   MOVEC0                                               00039013
000194 D203 C442 C4FA 00448 00500   181          MVC   BFROM(4),=X'00000000' from stack 0                   00039118
00019A D203 C446 C502 0044C 00508   182          MVC   BTO(4),=X'00000002' to stack 2                       00039218
0001A0 4140 0000      00000         183          LA    R4,0                next move is category R4 0       00039314
0001A4 47F0 C1B2      001B8         184          B     STPLPCNT                                             00039414
                                001A8 185 MOVEC0   EQU   *                category 0 move: 1-2 or vice versa 00039513
0001A8 D203 C442 C4FE 00448 00504   186          MVC   BFROM(4),=X'00000001' from stack 1                   00039618
0001AE D203 C446 C502 0044C 00508   187          MVC   BTO(4),=X'00000002' to stack 2                       00039718
0001B4 4140 0001      00001         188          LA    R4,1                next move is category 1          00039813
                                001B8 189 STPLPCNT EQU   *                                                 00039914
0001B8 47F0 C1C2      001C8         190          B     DOMOVE                                               00040014
                                    191 *                                                                  00040119
                                001BC 192 STPLPNXT EQU   *                                                 00040214
0001BC 5A30 C4F2      004F8         193          A     R3,=F'1'            increment steps                  00040319
0001C0 5030 C3BE      003C4         194          ST    R3,R3HELP                                            00040413
0001C4 47F0 C162      00168         195          B     STPLOOP                                              00040514
                                    196 *                                                                  00040614
                                001C8 197 DOMOVE   EQU   *                                                 00040714
                                    198 * Determine move direction (s to d or d to s) depending on what    00040814
                                    199 * is legal. Even if count(d) is 0 the comparison will work as      00040921
                                    200 * slice(stack,0) is initialized with a value higher than any number 00041021
                                    201 * of slices                                                       00041121
                                    202 * check STKSLICE(BFROM,STKCOUNT(BFROM)) < STKSLICE(BTO,STKCOUNT(BTO)) 00041221
                                    203 * if not change move direction                                    00041321
                                    204 * from side                                                       00041421
0001C8 4170 0004      00004         205          LA    R7,STKCOUNL                                          00041517
0001CC 5C60 C442      00448         206          M     R6,BFROM            STKCOUNTL*BFROM displacement     00041616
0001D0 4157 C3AA      003B0         207          LA    R5,STKCOUNT(R7)                                      00041719
0001D4 5050 C3CA      003D0         208          ST    R5,XFSTKCNT         address from stack count         00041821
0001D8 5897 C3AA      003B0         209          L     R9,STKCOUNT(R7)                                      00041919
0001DC 4150 0004      00004         210          LA    R5,STKDYNL                                           00042017
0001E0 1C85                         211          MR    R8,R5               relative BFROM count pos STKSLICE 00042121
0001E2 5870 C3B6      003BC         212          L     R7,STKSLCS1                                          00042216
0001E6 5C60 C442      00448         213          M     R6,BFROM            get stack position acc. to BFROM 00042317
0001EA 1A79                         214          AR    R7,R9               get STKSLICE position            00042417
0001EC 4157 B000      00000         215          LA    R5,STKSLICE(R7)                                      00042519
0001F0 5050 C3D2      003D8         216          ST    R5,XFSTKSLC         address from slice               00042619
0001F4 5867 B000      00000         217          L     R6,STKSLICE(R7)     get slice value                  00042717
0001F8 5060 C3C2      003C8         218          ST    R6,XSLCFROM         from slice value                 00042821
                                    219 * to side                                                          00042921
0001FC 4170 0004      00004         220          LA    R7,STKCOUNL                                          00043017
```

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  LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                        ASM 0201 20.50 11/01/26

000200 5C60 C446      0044C          221          M     R6,BTO              STKCOUNTL*BTO displacement          00043116
000204 4157 C3AA      003B0          222          LA    R5,STKCOUNT(R7)                                         00043219
000208 5050 C3CE      003D4          223          ST    R5,XTSTKCNT         address to stack count              00043319
00020C 5897 C3AA      003B0          224          L     R9,STKCOUNT(R7)                                         00043419
000210 4150 0004      00004          225          LA    R5,STKDYNL                                              00043517
000214 1C85                          226          MR    R8,R5               relative BTO count pos STKSLICE     00043621
000216 5870 C3B6      003BC          227          L     R7,STKSLCS1                                             00043716
00021A 5C60 C446      0044C          228          M     R6,BTO              get stack position acc. to BTO      00043821
00021E 1A79                          229          AR    R7,R9               get STKSLICE position               00043921
000220 4157 B000      00000          230          LA    R5,STKSLICE(R7)                                         00044019
000224 5050 C3D6      003DC          231          ST    R5,XTSTKSLC         address to slice                    00044119
000228 5867 B000      00000          232          L     R6,STKSLICE(R7)     get slice value                     00044221
00022C 5060 C3C6      003CC          233          ST    R6,XSLCTO           to slice value                      00044321
                                     234 *                                                                    00044421
000230 5960 C3C2      003C8          235          C     R6,XSLCFROM         compare to slice with from slice    00044519
000234 47B0 C27A      00280          236          BNL   CONTMOVE           top slc stk to >= top slc stk from  00044620
                                     237 * change direction of move as top slice to < top slice from          00044721
                                     238          SWAPF BFROM,BTO           swap from and to                  00044819
000238 D703 C442 C446 00448 0044C    239+         XC    BFROM,BTO                                               00025919
00023E D703 C446 C442 0044C 00448    240+         XC    BTO,BFROM                                               00026019
000244 D703 C442 C446 00448 0044C    241+         XC    BFROM,BTO                                               00026119
                                     242          SWAPF XFSTKCNT,XTSTKCNT   swap addresses from,to counts       00044919
00024A D703 C3CA C3CE 003D0 003D4    243+         XC    XFSTKCNT,XTSTKCNT                                       00025919
000250 D703 C3CE C3CA 003D4 003D0    244+         XC    XTSTKCNT,XFSTKCNT                                       00026019
000256 D703 C3CA C3CE 003D0 003D4    245+         XC    XFSTKCNT,XTSTKCNT                                       00026119
                                     246          SWAPF XFSTKSLC,XTSTKSLC   swap addresses from,to slice        00045019
00025C D703 C3D2 C3D6 003D8 003DC    247+         XC    XFSTKSLC,XTSTKSLC                                       00025919
000262 D703 C3D6 C3D2 003DC 003D8    248+         XC    XTSTKSLC,XFSTKSLC                                       00026019
000268 D703 C3D2 C3D6 003D8 003DC    249+         XC    XFSTKSLC,XTSTKSLC                                       00026119
                                     250          SWAPF XSLCFROM,XSLCTO     swap stored from and to slice values 00045119
00026E D703 C3C2 C3C6 003C8 003CC    251+         XC    XSLCFROM,XSLCTO                                         00025919
000274 D703 C3C6 C3C2 003CC 003C8    252+         XC    XSLCTO,XSLCFROM                                         00026019
00027A D703 C3C2 C3C6 003C8 003CC    253+         XC    XSLCFROM,XSLCTO                                         00026119
                                     254 * move the slice                                                     00045221
                           00280     255 CONTMOVE EQU   *                                                    00045318
000280 4150 0004      00004          256          LA    R5,STKDYNL          displacement for one slice element  00045419
000284 5A50 C3D6      003DC          257          A     R5,XTSTKSLC                                             00045520
000288 5860 C3C2      003C8          258          L     R6,XSLCFROM         from slice value                    00045619
00028C 5065 0000      00000          259          ST    R6,0(R5)            store from slice to to sclice+1     00045720
000290 5850 C3CE      003D4          260          L     R5,XTSTKCNT         to stack count                      00045820
000294 5865 0000      00000          261          L     R6,0(R5)                                                00045920
000298 5A60 C4F2      004F8          262          A     R6,=F'1'            add 1 to to stack count value       00046020
00029C 5065 0000      00000          263          ST    R6,0(R5)            store new value to to stack count   00046120
0002A0 5850 C3D2      003D8          264          L     R5,XFSTKSLC         set from slice to 0                 00046220
0002A4 4160 0000      00000          265          LA    R6,0                                                    00046319
0002A8 5065 0000      00000          266          ST    R6,0(R5)            set from slice to 0                 00046420
0002AC 5850 C3CA      003D0          267          L     R5,XFSTKCNT         from stack count                    00046520
0002B0 5865 0000      00000          268          L     R6,0(R5)                                                00046620
0002B4 5B60 C4F2      004F8          269          S     R6,=F'1'            subtract 1 from from stk count value 00046720
0002B8 5065 0000      00000          270          ST    R6,0(R5)            store new value to from stack count 00046820
                                     271 * prepare step information fields for output record                  00046921
0002BC 4E30 C3DA      003E0          272          CVD   R3,TPALIGN          current step                        00047019
0002C0 F837 C433 C3DA 00439 003E0    273          ZAP   STEPS,TPALIGN       store current step                  00047119
0002C6 5850 C3C2      003C8          274          L     R5,XSLCFROM         current slice                       00047219
0002CA 4E50 C3DA      003E0          275          CVD   R5,TPALIGN                                            00047319
```

```
  LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                          ASM 0201 20.50 11/01/26

0002CE F817 C42E C3DA 00434 003E0   276          ZAP   SLICES,TPALIGN      store current slice                  00047419
0002D4 4170 0001      00001         277          LA    R7,STKDSPNL                                              00047519
0002D8 5C60 C442      00448         278          M     R6,BFROM            STKDSPNL*BFROM displacement          00047619
0002DC 4190 C3A6      003AC         279          LA    R9,STKDSPNO                                              00047719
0002E0 1A97                         280          AR    R9,R7                                                    00047819
0002E2 F800 C430 9000 00436 00000   281          ZAP   FROM(1),0(1,R9)                                          00047919
0002E8 4170 0001      00001         282          LA    R7,STKDSPNL                                              00048019
0002EC 5C60 C446      0044C         283          M     R6,BTO              STKDSPNL*BFROM displacement          00048119
0002F0 4190 C3A6      003AC         284          LA    R9,STKDSPNO                                              00048219
0002F4 1A97                         285          AR    R9,R7                                                    00048319
0002F6 F800 C431 9000 00437 00000   286          ZAP   TO(1),0(1,R9)                                            00048419
                                    287 * prepare step line output record using the fields set above           00048521
0002FC D207 C4DB C4E3 004E1 004E9   288          MVC   TRGTSTPS,TMPSTPS                                         00048619
000302 DE07 C4DB C433 004E1 00439   289          ED    TRGTSTPS,STEPS                                           00048719
000308 D206 C450 C4DC 00456 004E2   290          MVC   OSTEP,TRGTSTPS+1                                         00048819
00030E D203 C4CF C4D3 004D5 004D9   291          MVC   TRGTSLC,TMPSLC                                           00048919
000314 DE03 C4CF C42E 004D5 00434   292          ED    TRGTSLC,SLICES                                           00049019
00031A D201 C466 C4D1 0046C 004D7   293          MVC   OSLICE,TRGTSLC+2                                         00049119
000320 D201 C4D7 C4D9 004DD 004DF   294          MVC   TRGTSTK,TMPSTK                                           00049219
000326 DE01 C4D7 C430 004DD 00436   295          ED    TRGTSTK,FROM                                             00049319
00032C D200 C474 C4D8 0047A 004DE   296          MVC   OFROM,TRGTSTK+1                                          00049419
000332 D201 C4D7 C4D9 004DD 004DF   297          MVC   TRGTSTK,TMPSTK                                           00049519
000338 DE01 C4D7 C431 004DD 00437   298          ED    TRGTSTK,TO                                               00049619
00033E D200 C479 C4D8 0047F 004DE   299          MVC   OTO,TRGTSTK+1                                            00049719
000344 5850 C42A      00430         300          L     R5,HANOIOUT         load address of HANOIOUT DCB         00049820
                                    301          PUT   (R5),OREC           write step line                      00049920
000348 1815                         302+         LR    1,R5                          LOAD PARAMETER REG 1   02050002
00034A 4100 C44A      00450         303+         LA    0,OREC                        LOAD PARAMETER REG 0   02500002
00034E 58F0 1030      00030         304+         L     15,48(0,1)              LOAD PUT ROUTINE ADDR        00550000
000352 05EF                         305+         BALR  14,15                   LINK TO PUT ROUTINE          00600000
000354 47F0 C1B6      001BC         306          B     STPLPNXT                                                 00050014
                                    307 *                                                                      00050119
                          00358     308 STPLPEND EQU   *                                                      00050214
000358 5B30 C4F2      004F8         309          S     R3,=F'1'                                                 00050319
00035C 4E30 C3DA      003E0         310          CVD   R3,TPALIGN                                               00050419
000360 F837 C433 C3DA 00439 003E0   311          ZAP   STEPS,TPALIGN       store number of steps performed      00050519
                                    312 * end of segment processing                                            00050601
                          00366     313 PROCEND  EQU   *                    end of segment processing            00050701
                                    314 * return to caller actions                                             00050801
000366 58D0 C3E6      003EC         315          L     R13,SAVEAREA+4                                           00050901
                                    316 * return current number of steps to caller (adress ASTEPS)             00051018
00036A 5830 C43A      00440         317          L     R3,ASTEPS                                                00051104
00036E F833 3000 C433 00000 00439   318          ZAP   0(4,R3),STEPS(4)                                         00051204
                                    319 * free dynamic stack data                                              00051305
000374 5800 C3BA      003C0         320          L     R0,STKALLOC         length of allocated  area            00051405
                                    321          FREEMAIN R,LV=(0),A=(11)  free area R11 with length R0       00051511
                                    322+*      OS/VS2 RELEASE 3 VERSION  -- 10/25/74                           00001603
000378 4110 B000      00000         323+         LA    1,0(0,11)                     LOAD AREA ADDRESS      00164002
00037C 0A0A                         324+         SVC   10                            ISSUE FREEMAIN SVC     00311202
                                    325          WTO   'HANOIITR... end HANOIITR'                             00051605
00037E 0700                         326+         CNOP  0,4                                                    09800002
000380 4510 C39A      003A0         327+         BAL   1,IHB0012A                    BRANCH AROUND MESSAGE  09850002
000384 001C                         328+         DC    AL2(28)             TEXT LENGTH                        13200002
000386 0000                         329+         DC    B'0000000000000000' MCS FLAGS                          13250002
000388 C8C1D5D6C9C9E3D9             330+         DC    C'HANOIITR... end HANOIITR'                            13350002
```

```
 LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                  ASM 0201 20.50 11/01/26

0003A0                            331+IHB0012A DS    0H                                                      15850002
0003A0 0A23                       332+         SVC   35                                                      15950002
                                  333          RETURN (14,12),RC=0                                     00051701
0003A2 98EC D00C      0000C       334+         LM    14,12,12(13)                RESTORE THE REGISTERS       00650000
0003A6 41F0 0000      00000       335+         LA    15,0(0,0)                   LOAD RETURN CODE            01750000
0003AA 07FE                       336+         BR    14                          RETURN                      02000000
                                  337 *---                                                                  00051821
                                  338 * debug                                                               00051917
                                  339 *        L     R5,BSLICES                                             00052017
                                  340 *        CVD   R5,TPALIGN                                             00052117
                                  341 *        MVC   TRGTVAL,TMPVAL                                         00052219
                                  342 *        ED    TRGTVAL,TPALIGN                                        00052319
                                  343 *        MVC   MSGVALUE,TRGTVAL                                       00052419
                                  344 *        WTO   MF=(E,MSGBLOCK)                                        00052517
                                  345 *        B     PROCEND                                                00052617
                                  346 *                                                                     00052717
                                  347 *                                                                     00052819
                                  348 *--- data definitions                                                 00052901
                                  349 * info for the 3 stacks                                              00053005
0003AC                            350 STKDSPNO DS    3PL1                display no of the stack             00053105
                         00001    351 STKDSPNL EQU   1                                                     00053217
0003B0                            352 STKCOUNT DS    3F                  count                               00053315
                         00004    353 STKCOUNL EQU   4                                                     00053417
                                  354 *--- stack slice info - dynamically depending on no of slices         00053521
                                  355 * allocate 3 times no of slices (3 times no of slices STKDYNL bytes)  00053605
0003BC                            356          CNOP  0,4                                                    00053711
000000                            357 STKDYN   DSECT                                                        00053805
000000                            358 STKSLICE DS    XL4                                                    00053911
                         00004    359 STKDYNL  EQU   *-STKDYN                                               00054005
                                  360 * end DSECT, continue CSECT                                           00054105
0003BC                            361 HANOIITR CSECT                                                        00054205
0003BC                            362          CNOP  0,4                                                    00054305
0003BC                            363 STKSLCS1 DS    F                   dnynamic slices+1 storage per stack 00054411
0003C0                            364 STKALLOC DS    F                   length of alloced dyn stack area    00054511
0003C4                            365 R3HELP   DS    F                   R3 value in storage as help         00054618
0003C8                            366 XSLCFROM DS    F                   number of slice to move from        00054718
0003CC                            367 XSLCTO   DS    F                   number of slice to move to          00054818
0003D0                            368 XFSTKCNT DS    F                   address from stack count            00054919
0003D4                            369 XTSTKCNT DS    F                   address to stack count              00055019
0003D8                            370 XFSTKSLC DS    F                   address from slice                  00055119
0003DC                            371 XTSTKSLC DS    F                   address to slice                    00055219
0003E0                            372          CNOP  0,8                                                    00055305
0003E0                            373 TPALIGN  DS    PL8                 aligned packed decimal 8 bytes long 00055409
                                  374 *---                                                                  00055521
0003E8                            375 SAVEAREA DS    18F                                                    00055601
                                  376 *--- parameters (by value except reference to output file DCB HANOIOUT) 00055721
                                  377 * remember address of variable (reference, output) for STEPS in ASTEPS 00055804
                                  378 * to return number of steps to caller                                 00055904
000430                            379 HANOIOUT DS    A(4)                address of output file DCB          00056001
000434                            380 SLICES   DS    PL2                                                    00056104
000436                            381 FROM     DS    PL1                                                    00056201
000437                            382 TO       DS    PL1                                                    00056301
000438                            383 AUX      DS    PL1                                                    00056401
000439                            384 STEPS    DS    PL4                                                    00056501
000440                            385 ASTEPS   DS    A(4)                remember caller's steps address     00056604
```

```
  LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                ASM 0201 20.50 11/01/26

                                    386 * parameters a binary values                                          00056710
000444                              387 BSLICES  DS    F                                                      00056811
000448                              388 BFROM    DS    F                                                      00056911
00044C                              389 BTO      DS    F                                                      00057011
                                    390 *--- output line definition for step                                  00057121
000450                              391 OREC     DS    0CL133                                                 00057201
000450 40                           392          DC    CL1' '              ASO character                      00057301
000451 E2A3859740                   393          DC    CL5'Step '                                             00057401
000456                              394 OSTEP    DS    CL7                 current step number                00057501
00045D 7A40D496A5899587             395          DC    CL15': Moving slice '                                  00057601
00046C                              396 OSLICE   DS    CL2                 current slice                      00057701
00046E 408699969440A2A3             397          DC    CL12' from stack '                                     00057801
00047A                              398 OFROM    DS    CL1                 from stack                         00057901
00047B 40A39640                     399          DC    CL4' to '                                              00058001
00047F                              400 OTO      DS    CL1                 to stack                           00058101
000480 4040404040404040             401          DC    CL85' '                                                00058201
                                    402 * templates and target fields for print preparation                    00058301
0004D5                              403 TRGTSLC  DS    CL4                                                    00058401
0004D9 40202120                     404 TMPSLC   DC    X'40202120'                                            00058519
0004DD                              405 TRGTSTK  DS    CL2                                                    00058601
0004DF 4020                         406 TMPSTK   DC    X'4020'                                                00058719
0004E1                              407 TRGTSTPS DS    CL8                                                    00058801
0004E9 4020202020202120             408 TMPSTPS  DC    X'4020202020202120'                                    00058919
                                    409 *---                                                                  00059021
                                    410 *                                                                     00059119
                                    411 * WTO message for debug purposes                                      00059207
                                    412 *         CNOP  0,4                                                   00059320
                                    413 *MSGBLOCK EQU   *                                                     00059420
                                    414 *         DC    H'80'               length of WTO buffer              00059520
                                    415 *         DC    H'0'                should be binary zeroes           00059620
                                    416 *MSGTEXT  DS    0CL76                                                 00059720
                                    417 *         DC    CL11'HANOIITR...'                                     00059820
                                    418 *MSGVALUE DS    CL16                                                  00059920
                                    419 *         DC    CL3'...'                                              00060020
                                    420 *         DC    CL46' '                                               00060120
                                    421 * WTO msg additional stuff                                           00060220
                                    422 *TRGTVAL  DS    CL16                                                  00060320
                                    423 *TMPVAL   DC    X'40202020202020202020202020202120'                   00060420
                                    424 *                                                                     00060519
                                    425 *                                                                     00060619
                                    426 *--- literals, if any, will go here                                   00060701
0004F8                              427          LTORG *                                                      00060801
0004F8 00000001                     428                =F'1'
0004FC 00000003                     429                =F'3'
000500 00000000                     430                =X'00000000'
000504 00000001                     431                =X'00000001'
000508 00000002                     432                =X'00000002'
                                    433 *--- end of section                                                  00060901
000000                              434          END   HANOIITR                                               00061001
```

SYMBOL	LEN	VALUE	DEFN	REFERENCES
ASTEPS	00004	00000440	00385	00076 00317
AUX	00001	00000438	00383	00073 00109 00112
BFROM	00004	00000448	00388	00175 00181 00186 00206 00213 00239 00240 00241 00278
BSLICES	00004	00000444	00387	00080 00083 00105 00118 00135 00162
BTO	00004	0000044C	00389	00176 00182 00187 00221 00228 00239 00240 00241 00283
CONTEVEN	00001	000000D2	00111	00107
CONTINIT	00001	000000DE	00114	00110
CONTMOVE	00001	00000280	00255	00236
DOMOVE	00001	000001C8	00197	00190
ENDINIT	00001	00000150	00159	00142
FROM	00001	00000436	00381	00069 00104 00281 00295
HANOIITR	00001	00000000	00043	00361 00434
HANOIOUT	00004	00000430	00379	00065 00300
IHB0003A	00002	0000003A	00061	00057
IHB0012A	00002	000003A0	00331	00327
MOVEC0	00001	000001A8	00185	00180
MOVEC2	00004	0000018C	00179	00174
OFROM	00001	0000047A	00398	00296
OREC	00133	00000450	00391	00303
OSLICE	00002	0000046C	00396	00293
OSTEP	00007	00000456	00394	00290
OTO	00001	0000047F	00400	00299
R0	00001	00000000	00027	00320
R1	00001	00000001	00028	00049 00095
R11	00001	0000000B	00038	00095 00096
R12	00001	0000000C	00039	00047 00048
R13	00001	0000000D	00040	00051 00052 00053 00315
R2	00001	00000002	00029	00049 00063 00086 00088 00128 00129 00131 00133 00135 00137 00141 00154 00161 00163 00171
R3	00001	00000003	00030	00050 00051 00053 00063 00064 00066 00068 00070 00072 00074 00079 00080 00083 00084 00087 00089 00130 00131 00132 00133 00136 00144 00155 00162 00163 00164 00165 00193 00194 00272 00309 00310 00317 00318
R3HELP	00004	000003C4	00365	00165 00171 00194
R4	00001	00000004	00031	00064 00065 00066 00067 00068 00069 00070 00071 00072 00073 00074 00075 00076 00085 00086 00139 00140 00150 00152 00170 00173 00177 00179 00183 00188
R5	00001	00000005	00032	00105 00106 00137 00144 00148 00157 00207 00208 00210 00211 00215 00216 00222 00223 00225 00226 00230 00231 00256 00257 00259 00260 00261 00263 00264 00266 00267 00268 00270 00274 00275 00300 00302
R6	00001	00000006	00033	00138 00139 00147 00148 00149 00150 00151 00152 00156 00157 00206 00213 00217 00218 00221 00228 00232 00233 00235 00258 00259 00261 00262 00263 00265 00266 00268 00269 00270 00278 00283
R7	00001	00000007	00034	00205 00207 00209 00212 00214 00215 00217 00220 00222 00224 00227 00229 00230 00232 00277 00280 00282 00285
R8	00001	00000008	00035	00211 00226
R9	00001	00000009	00036	00209 00214 00224 00229 00279 00280 00281 00284 00285 00286
SAVEAREA	00004	000003E8	00375	00050 00052 00315
SLCINIT	00004	0000011C	00141	00158
SLICES	00002	00000434	00380	00067 00078 00276 00292
STEPS	00004	00000439	00384	00075 00273 00289 00311 00318
STKALLOC	00004	000003C0	00364	00089 00320
STKCOUNL	00001	00000004	00353	00119 00120 00205 00220
STKCOUNT	00004	000003B0	00352	00118 00119 00120 00207 00209 00222 00224
STKDSPNL	00001	00000001	00351	00108 00109 00112 00113 00277 00282
STKDSPNO	00001	000003AC	00350	00104 00108 00109 00112 00113 00279 00284
STKDYN	00001	00000000	00357	00096 00359
STKDYNL	00001	00000004	00359	00085 00138 00156 00210 00225 00256

```
SYMBOL    LEN   VALUE   DEFN    REFERENCES

STKSLCS1 00004 000003BC 00363  00087 00130 00132 00149 00151 00212 00227
STKSLICE 00004 00000000 00358  00129 00131 00133 00144 00147 00215 00217 00230 00232
STPLOOP  00004 00000168 00171  00195
STPLPCNT 00001 000001B8 00189  00178 00184
STPLPEND 00001 00000358 00308  00172
STPLPNXT 00001 000001BC 00192  00306
TMPSLC   00004 000004D9 00404  00291
TMPSTK   00002 000004DF 00406  00294 00297
TMPSTPS  00008 000004E9 00408  00288
TO       00001 00000437 00382  00071 00108 00113 00286 00298
TPALIGN  00008 000003E0 00373  00078 00079 00272 00273 00275 00276 00310 00311
TRGTSLC  00004 000004D5 00403  00291 00292 00293
TRGTSTK  00002 000004DD 00405  00294 00295 00296 00297 00298 00299
TRGTSTPS 00008 000004E1 00407  00288 00289 00290
XFSTKCNT 00004 000003D0 00368  00208 00243 00244 00245 00267
XFSTKSLC 00004 000003D8 00370  00216 00247 00248 00249 00264
XSLCFROM 00004 000003C8 00366  00218 00235 00251 00252 00253 00258 00274
XSLCTO   00004 000003CC 00367  00233 00251 00252 00253
XTSTKCNT 00004 000003D4 00369  00223 00243 00244 00245 00260
XTSTKSLC 00004 000003DC 00371  00231 00247 00248 00249 00257
```

```
SYMBOL    LEN   VALUE   DEFN    REFERENCES

=F'1'     00004 000004F8 00428  00084 00106 00136 00154 00155 00193 00262 00269 00309
=F'3'     00004 000004FC 00429  00088
=X'00000000'
          00004 00000500 00430  00119 00120 00175 00181
=X'00000001'
          00004 00000504 00431  00141 00173 00176 00186
=X'00000002'
          00004 00000508 00432  00179 00182 00187
```

```
NO STATEMENTS FLAGGED IN THIS ASSEMBLY
HIGHEST SEVERITY WAS    0
OPTIONS FOR THIS ASSEMBLY
  ALIGN, ALOGIC, BUFSIZE(STD), NODECK, ESD, FLAG(0), LINECOUNT(55), LIST, NOMCALL, YFLAG, WORKSIZE(2097152)
  NOMLOGIC, NONUMBER, OBJECT, NORENT, RLD, NOSTMT, NOLIBMAC, NOTERMINAL, NOTEST, XREF(SHORT)
  SYSPARM()
WORK FILE BUFFER SIZE/NUMBER = 8702/ 1
TOTAL RECORDS READ FROM SYSTEM INPUT        371
TOTAL RECORDS READ FROM SYSTEM LIBRARY     2004
TOTAL RECORDS PUNCHED                        29
TOTAL RECORDS PRINTED                       556
```

```
SYMBOL   TYPE  ID   ADDR  LENGTH LDID                                    ASM 0201 20.50 11/01/26

          PC  0001 000000 000000
HANOIASM  SD  0002 000000 000898
HANOIITR  ER  0003
HANOIREC  ER  0004
```

```
LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                ASM 0201 20.50 11/01/26

                                    1 ********************************************************************** 00061601
                                    2 * ////////////////////////////////////////////////////////////////// 00061701
                                    3 * Title: HANOIASM - The Towers of Hanoi                               00061801
                                    4 * Author: ManInBlueAT                                                 00061901
                                    5 * Language: ASM                                                       00062001
                                    6 *                                                                     00062101
                                    7 * Description: Move n slices from source stack from to destination    00062221
                                    8 *   stack to using auxiliary stack aux (6-from-to). A recursive or    00062321
                                    9 *   iterative algorithm is applied according to the choice (REC or    00062421
                                   10 *   ITER) entered in the input record. Please refer to                00062521
                                   11 *   https://en.wikipedia.org/wiki/Tower_of_Hanoi                      00062621
                                   12 *                                                                     00062701
                                   13 * Changes:                                                            00062801
                                   14 * - ManInBlueAT, 2026-10-09 New program                              00062901
                                   15 * ////////////////////////////////////////////////////////////////// 00063001
                                   16          PRINT GEN                                                   00063101
                                   17          REGISTER                                                    00063201
                        00000      18+R0       EQU   0                          , GENERAL PURPOSE REGISTER 0
                        00001      19+R1       EQU   1                          , GENERAL PURPOSE REGISTER 1
                        00002      20+R2       EQU   2                          , GENERAL PURPOSE REGISTER 2
                        00003      21+R3       EQU   3                          , GENERAL PURPOSE REGISTER 3
                        00004      22+R4       EQU   4                          , GENERAL PURPOSE REGISTER 4
                        00005      23+R5       EQU   5                          , GENERAL PURPOSE REGISTER 5
                        00006      24+R6       EQU   6                          , GENERAL PURPOSE REGISTER 6
                        00007      25+R7       EQU   7                          , GENERAL PURPOSE REGISTER 7
                        00008      26+R8       EQU   8                          , GENERAL PURPOSE REGISTER 8
                        00009      27+R9       EQU   9                          , GENERAL PURPOSE REGISTER 9
                        0000A      28+R10      EQU   10                         , GENERAL PURPOSE REGISTER 10
                        0000B      29+R11      EQU   11                         , GENERAL PURPOSE REGISTER 11
                        0000C      30+R12      EQU   12                         , GENERAL PURPOSE REGISTER 12
                        0000D      31+R13      EQU   13                         , GENERAL PURPOSE REGISTER 13
                        0000E      32+R14      EQU   14                         , GENERAL PURPOSE REGISTER 14
                        0000F      33+R15      EQU   15                         , GENERAL PURPOSE REGISTER 15
000000                             34 HANOIASM CSECT                                                       00063301
                                   35          SAVE  (14,12)                                               00063401
000000                             36+         DS    0H                                                    01650000
000000 90EC D00C      0000C        37+         STM   14,12,12(13)                    SAVE REGISTERS        02950000
000004 05C0                        38          BALR  R12,0                                                 00063501
                        00006      39          USING *,R12                                                 00063601
000006 4120 C39E      003A4        40          LA    R2,SAVEAREA                                           00063701
00000A 5020 D008      00008        41          ST    R2,8(,R13)                                            00063801
00000E 50D0 C3A2      003A8        42          ST    R13,SAVEAREA+4                                        00063901
000012 18D2                        43          LR    R13,R2                                                00064001
                                   44 *--- start processing                                                00064101
                                   45          WTO   'HANOIASM... Begin execution'                         00064201
000014                             46+         CNOP  0,4                                                   09800002
000014 4510 C032      00038        47+         BAL   1,IHB0003A                      BRANCH AROUND MESSAGE 09850002
000018 001F                        48+         DC    AL2(31)             TEXT LENGTH                       13200002
00001A 0000                        49+         DC    B'0000000000000000' MCS FLAGS                         13250002
00001C C8C1D5D6C9C1E2D4            50+         DC    C'HANOIASM... Begin execution'                        13350002
000038                             51+IHB0003A DS    0H                                                    15850002
000038 0A23                        52+         SVC   35                                                    15950002
00003A 41B0 0000      00000        53          LA    R11,0               return code 0                     00064301
                                   54          OPEN  (HANOIIN,(INPUT))                                     00064401
00003E 0700                        55+         CNOP  0,4                             ALIGN LIST TO FULLWORD 01740001
```

```
  LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                  ASM 0201 20.50 11/01/26

000040 4510 C042      00048          56+         BAL   1,*+8                          LOAD REG1 W/LIST ADDR. 01780000
000044 80                            57+         DC    AL1(128)                       OPTION BYTE            01900000
000045 000414                        58+         DC    AL3(HANOIIN)                   DCB ADDRESS            01920000
000048 0A13                          59+         SVC   19                             ISSUE OPEN SVC         04000000
                                     60          OPEN  (HANOIOUT,(OUTPUT))                                   00064501
00004A 0700                          61+         CNOP  0,4                            ALIGN LIST TO FULLWORD 01740001
00004C 4510 C04E      00054          62+         BAL   1,*+8                          LOAD REG1 W/LIST ADDR. 01780000
000050 8F                            63+         DC    AL1(143)                       OPTION BYTE            01900000
000051 000474                        64+         DC    AL3(HANOIOUT)                  DCB ADDRESS            01920000
000054 0A13                          65+         SVC   19                             ISSUE OPEN SVC         04000000
                                     66 * read towers of hanoi runtime parameters: slices from to               00064601
                                     67          GET   HANOIIN,IREC                                          00064701
000056 4110 C40E      00414          68+         LA    1,HANOIIN                      LOAD PARAMETER REG 1   01900002
00005A 4100 C4CE      004D4          69+         LA    0,IREC                         LOAD PARAMETER REG 0   02500002
00005E 58F0 1030      00030          70+         L     15,48(0,1)               LOAD GET ROUTINE ADDR        00600000
000062 05EF                          71+         BALR  14,15                    LINK TO GET ROUTINE          00625000
                                     72 * check parameters                                                   00064801
000064 F211 C3EA C4CF 003F0 004D5    73          PACK  SLICES,ISLICES                                        00064901
00006A F910 C3EA C792 003F0 00798    74          CP    SLICES,=P'0'                                          00065001
000070 47D0 C286      0028C          75          BNP   ERRP1                  <=0 or overlfow BC X'0D'       00065101
000074 F911 C3EA C76A 003F0 00770    76          CP    SLICES,=P'20'                                         00065201
00007A 4720 C286      0028C          77          BP    ERRP1                  >11 BC X'02'                   00065301
                                     78 *                                                                    00065401
00007E F200 C3EC C4D2 003F2 004D8    79          PACK  FROM,IFROM                                            00065501
000084 F900 C3EC C792 003F2 00798    80          CP    FROM,=P'0'                                            00065601
00008A 47D0 C2B4      002BA          81          BNP   ERRP23                 <=0 or overflow BC X'0D'       00065701
00008E F900 C3EC C793 003F2 00799    82          CP    FROM,=P'3'                                            00065801
000094 4720 C2B4      002BA          83          BP    ERRP23                 >3 BC X'02'                    00065901
                                     84 *                                                                    00066001
000098 F200 C3ED C4D4 003F3 004DA    85          PACK  TO,ITO                                                00066101
00009E F900 C3ED C792 003F3 00798    86          CP    TO,=P'0'                                              00066201
0000A4 47D0 C2B4      002BA          87          BNP   ERRP23                 <=0 or overflow BC X'0D'       00066301
0000A8 F900 C3ED C793 003F3 00799    88          CP    TO,=P'3'                                              00066401
0000AE 4720 C2B4      002BA          89          BP    ERRP23                 >3 BC X'02'                    00066501
0000B2 F900 C3EC C3ED 003F2 003F3    90          CP    FROM,TO                                               00066601
0000B8 4780 C2B4      002BA          91          BZ    ERRP23                 = BC X'08'                     00066701
                                     92 *                                                                    00066801
0000BC D503 C4D6 C752 004DC 00758    93          CLC   IALG,=C'REC '                                         00066901
0000C2 4780 C0CA      000D0          94          BE    C4OK                   = BC X'08'                     00067001
0000C6 D503 C4D6 C756 004DC 0075C    95          CLC   IALG,=C'ITER'                                         00067101
0000CC 4770 C2E2      002E8          96          BNE   ERRP4                  = BC X'07'                     00067201
                           000D0     97 C4OK     EQU   *                                                     00067301
                                     98 * all checks ok, determine AUX stack number                          00067401
0000D0 F800 C3EE C794 003F4 0079A    99          ZAP   AUX,=P'6'              aux=6-from-to                  00067501
0000D6 FB00 C3EE C3EC 003F4 003F2   100          SP    AUX,FROM                                              00067601
0000DC FB00 C3EE C3ED 003F4 003F3   101          SP    AUX,TO                                                00067701
                                    102 * prepare headline OHEAD                                             00067801
0000E2 D203 C628 C62C 0062E 00632   103          MVC   TRGTSLC,TMPSLC                                        00067901
0000E8 DE03 C628 C3EA 0062E 003F0   104          ED    TRGTSLC,SLICES                                        00068001
0000EE D201 C526 C62A 0052C 00630   105          MVC   OHSLICES,TRGTSLC+2                                    00068101
0000F4 D201 C630 C632 00636 00638   106          MVC   TRGTSTK,TMPSTK                                        00068201
0000FA DE01 C630 C3EC 00636 003F2   107          ED    TRGTSTK,FROM                                          00068301
000100 D200 C542 C631 00548 00637   108          MVC   OHFROM,TRGTSTK+1                                      00068401
000106 D201 C630 C632 00636 00638   109          MVC   TRGTSTK,TMPSTK                                        00068501
00010C DE01 C630 C3ED 00636 003F3   110          ED    TRGTSTK,TO                                            00068601
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  LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                ASM 0201 20.50 11/01/26

000112 D200 C559 C631 0055F 00637   111          MVC   OHTO,TRGTSTK+1                                        00068701
000118 D201 C630 C632 00636 00638   112          MVC   TRGTSTK,TMPSTK                                        00068801
00011E DE01 C630 C3EE 00636 003F4   113          ED    TRGTSTK,AUX                                           00068901
000124 D200 C571 C631 00577 00637   114          MVC   OHAUX,TRGTSTK+1                                       00069001
                                    115          PUT   HANOIOUT,OHEAD      write headline                    00069101
00012A 4110 C46E      00474         116+         LA    1,HANOIOUT                    LOAD PARAMETER REG 1    01900002
00012E 4100 C51E      00524         117+         LA    0,OHEAD                       LOAD PARAMETER REG 0    02500002
000132 58F0 1030      00030         118+         L     15,48(0,1)              LOAD PUT ROUTINE ADDR         00550000
000136 05EF                         119+         BALR  14,15                   LINK TO PUT ROUTINE           00600000
                                    120          PUT   HANOIOUT,=CL133' '  write space line                00069201
000138 4110 C46E      00474         121+         LA    1,HANOIOUT                    LOAD PARAMETER REG 1    01900002
00013C 4100 C795      0079B         122+         LA    0,=CL133' '                   LOAD PARAMETER REG 0    02500002
000140 58F0 1030      00030         123+         L     15,48(0,1)              LOAD PUT ROUTINE ADDR         00550000
000144 05EF                         124+         BALR  14,15                   LINK TO PUT ROUTINE           00600000
                                    125 * initialize number of steps                                        00069301
000146 F830 C3EF C792 003F5 00798   126          ZAP   STEPS,=P'0'                                           00069401
                                    127 * call HANOILN to peform the necessary steps                        00069501
00014C D503 C4D6 C752 004DC 00758   128          CLC   IALG,=C'REC '                                         00069601
000152 4780 C1BE      001C4         129          BE    CALLREC             = BC X'08'                        00069701
                                    130          WTO   'HANOIASM... calling HANOIITR'                        00069801
000156 0700                         131+         CNOP  0,4                                                   09800002
000158 4510 C176      0017C         132+         BAL   1,IHB0012A                    BRANCH AROUND MESSAGE   09850002
00015C 0020                         133+         DC    AL2(32)             TEXT LENGTH                       13200002
00015E 0000                         134+         DC    B'0000000000000000' MCS FLAGS                         13250002
000160 C8C1D5D6C9C1E2D4             135+         DC    C'HANOIASM... calling HANOIITR'                       13350002
00017C                              136+IHB0012A DS    0H                                                    15850002
00017C 0A23                         137+         SVC   35                                                    15950002
                                    138          CALL  HANOIITR,(ALIST)                                      00069901
00017E 0700                         139+         CNOP  0,4                                                   00750000
000180 47F0 C182      00188         140+         B     *+8                     BRANCH AROUND VCON            00800000
000184 00000000                     141+IHB0013B DC    V(HANOIITR)             ENTRY POINT ADDRESS           00850000
000188                              142+         CNOP  0,4                                                   02800000
000188 4510 C18A      00190         143+         BAL   1,IHB0014A                    LOAD LIST ADDR IN REG1  02850000
                            0018C   144+IHB0014  EQU   *                                                     02900000
00018C 00                           145+         DC    B'00000000'               SET VL SWITCH BIT         03700000
00018D 0003FC                       146+         DC    AL3(ALIST)                PROB. PROG. PARAMETER     03750000
                            00190   147+IHB0014A EQU   *                                                     04100000
000190 58F0 C17E      00184         148+         L     15,IHB0013B               LOAD 15 WITH ENTRY ADR    01100000
000194 05EF                         149+         BALR  14,15                     BRANCH TO ENTRY POINT     01150000
                                    150          WTO   'HANOIASM... back from HANOIITR'                      00070001
000196 0700                         151+         CNOP  0,4                                                   09800002
000198 4510 C1B8      001BE         152+         BAL   1,IHB0015A                    BRANCH AROUND MESSAGE   09850002
00019C 0022                         153+         DC    AL2(34)             TEXT LENGTH                       13200002
00019E 0000                         154+         DC    B'0000000000000000' MCS FLAGS                         13250002
0001A0 C8C1D5D6C9C1E2D4             155+         DC    C'HANOIASM... back from HANOIITR'                     13350002
0001BE                              156+IHB0015A DS    0H                                                    15850002
0001BE 0A23                         157+         SVC   35                                                    15950002
0001C0 47F0 C226      0022C         158          B     PREPEND                                               00070101
                            001C4   159 CALLREC  EQU   *                                                     00070201
                                    160          WTO   'HANOIASM... calling HANOIREC'                        00070301
0001C4                              161+         CNOP  0,4                                                   09800002
0001C4 4510 C1E2      001E8         162+         BAL   1,IHB0016A                    BRANCH AROUND MESSAGE   09850002
0001C8 0020                         163+         DC    AL2(32)             TEXT LENGTH                       13200002
0001CA 0000                         164+         DC    B'0000000000000000' MCS FLAGS                         13250002
0001CC C8C1D5D6C9C1E2D4             165+         DC    C'HANOIASM... calling HANOIREC'                       13350002
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  LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                  ASM 0201 20.50 11/01/26

0001E8                            166+IHB0016A DS    0H                                                        15850002
0001E8 0A23                       167+         SVC   35                                                        15950002
                                  168          CALL  HANOIREC,(ALIST)                                          00070401
0001EA 0700                       169+         CNOP  0,4                                                       00750000
0001EC 47F0 C1EE      001F4       170+         B     *+8                 BRANCH AROUND VCON                    00800000
0001F0 00000000                   171+IHB0017B DC    V(HANOIREC)         ENTRY POINT ADDRESS                   00850000
0001F4                            172+         CNOP  0,4                                                       02800000
0001F4 4510 C1F6      001FC       173+         BAL   1,IHB0018A          LOAD LIST ADDR IN REG1                02850000
                            001F8 174+IHB0018  EQU   *                                                         02900000
0001F8 00                         175+         DC    B'00000000'         SET VL SWITCH BIT                     03700000
0001F9 0003FC                     176+         DC    AL3(ALIST)          PROB. PROG. PARAMETER                 03750000
                            001FC 177+IHB0018A EQU   *                                                         04100000
0001FC 58F0 C1EA      001F0       178+         L     15,IHB0017B         LOAD 15 WITH ENTRY ADR                01100000
000200 05EF                       179+         BALR  14,15               BRANCH TO ENTRY POINT                 01150000
                                  180          WTO   'HANOIASM... back from HANOIREC'                          00070501
000202 0700                       181+         CNOP  0,4                                                       09800002
000204 4510 C224      0022A       182+         BAL   1,IHB0019A          BRANCH AROUND MESSAGE                 09850002
000208 0022                       183+         DC    AL2(34)               TEXT LENGTH                         13200002
00020A 0000                       184+         DC    B'0000000000000000' MCS FLAGS                           13250002
00020C C8C1D5D6C9C1E2D4           185+         DC    C'HANOIASM... back from HANOIREC'                       13350002
00022A                            186+IHB0019A DS    0H                                                        15850002
00022A 0A23                       187+         SVC   35                                                        15950002
                                  188 * prepare trailing line OTRAIL                                            00070601
                            0022C 189 PREPEND  EQU   *                                                         00070701
00022C D207 C634 C63C 0063A 00642 190          MVC   TRGTSTPS,TMPSTPS                                          00070801
000232 DE07 C634 C3EF 0063A 003F5 191          ED    TRGTSTPS,STEPS                                            00070901
000238 D206 C5AC C635 005B2 0063B 192          MVC   OTSTEPS,TRGTSTPS+1                                        00071001
                                  193          PUT   HANOIOUT,=CL133' '  write space line                      00071101
00023E 4110 C46E      00474       194+         LA    1,HANOIOUT                  LOAD PARAMETER REG 1          01900002
000242 4100 C795      0079B       195+         LA    0,=CL133' '                 LOAD PARAMETER REG 0          02500002
000246 58F0 1030      00030       196+         L     15,48(0,1)            LOAD PUT ROUTINE ADDR               00550000
00024A 05EF                       197+         BALR  14,15                 LINK TO PUT ROUTINE               00600000
                                  198          PUT   HANOIOUT,OTRAIL     write trailing line                   00071201
00024C 4110 C46E      00474       199+         LA    1,HANOIOUT                  LOAD PARAMETER REG 1          01900002
000250 4100 C5A3      005A9       200+         LA    0,OTRAIL                    LOAD PARAMETER REG 0          02500002
000254 58F0 1030      00030       201+         L     15,48(0,1)            LOAD PUT ROUTINE ADDR               00550000
000258 05EF                       202+         BALR  14,15                 LINK TO PUT ROUTINE               00600000
00025A 47F0 C310      00316       203          B     PROCEND             end of program                        00071301
                                  204 * handle no input record provided                                         00071401
                            0025E 205 ATFIEND  EQU   *                                                         00071501
00025E D203 C6CD C75A 006D3 00760 206          MVC   OMSGNO,=C'0001'                                           00071601
000264 D200 C6D1 C81A 006D7 00820 207          MVC   OMSGTYPE,=C'E'                                            00071701
00026A D27A C6D3 C644 006D9 0064A 208          MVC   OMSGTEXT,BLANK                                            00071801
000270 D214 C6D3 C81B 006D9 00821 209          MVC   OMSGTEXT(21),=C'No input record found'                    00071901
                                  210          PUT   HANOIOUT,OMSG                                             00072001
000276 4110 C46E      00474       211+         LA    1,HANOIOUT                  LOAD PARAMETER REG 1          01900002
00027A 4100 C6C9      006CF       212+         LA    0,OMSG                      LOAD PARAMETER REG 0          02500002
00027E 58F0 1030      00030       213+         L     15,48(0,1)            LOAD PUT ROUTINE ADDR               00550000
000282 05EF                       214+         BALR  14,15                 LINK TO PUT ROUTINE               00600000
000284 41B0 0008      00008       215          LA    R11,8                 return code                         00072101
000288 47F0 C310      00316       216          B     PROCEND             end of program                        00072201
                                  217 * parameter error messages                                                00072301
                            0028C 218 ERRP1    EQU   *                                                         00072401
00028C D203 C6CD C75E 006D3 00764 219          MVC   OMSGNO,=C'0002'                                           00072501
000292 D200 C6D1 C81A 006D7 00820 220          MVC   OMSGTYPE,=C'E'                                            00072601
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  LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                  ASM 0201 20.50 11/01/26

000298 D27A C6D3 C644 006D9 0064A   221          MVC   OMSGTEXT,BLANK                                        00072701
00029E D225 C6D3 C76C 006D9 00772   222          MVC   OMSGTEXT(38),=C'Number of slices must be 1<=slices<=20'  00072801
                                    223          PUT   HANOIOUT,OMSG       Write error message               00072901
0002A4 4110 C46E      00474         224+         LA    1,HANOIOUT                    LOAD PARAMETER REG 1   01900002
0002A8 4100 C6C9      006CF         225+         LA    0,OMSG                        LOAD PARAMETER REG 0   02500002
0002AC 58F0 1030      00030         226+         L     15,48(0,1)              LOAD PUT ROUTINE ADDR        00550000
0002B0 05EF                         227+         BALR  14,15                   LINK TO PUT ROUTINE          00600000
0002B2 41B0 0008      00008         228          LA    R11,8                 return code                    00073001
0002B6 47F0 C310      00316         229          B     PROCEND               end of program                 00073101
                                    230 *                                                                    00073201
                            002BA   231 ERRP23   EQU   *                                                    00073301
0002BA D203 C6CD C762 006D3 00768   232          MVC   OMSGNO,=C'0003'                                       00073401
0002C0 D200 C6D1 C81A 006D7 00820   233          MVC   OMSGTYPE,=C'E'                                        00073501
0002C6 D27A C6D3 C644 006D9 0064A   234          MVC   OMSGTEXT,BLANK                                        00073601
0002CC D229 C6D3 C830 006D9 00836   235          MVC   OMSGTEXT(42),=C'Values from,to must be 1<=from,to<=3 andX00073701
                                                        from<>to'                                           00073801
                                    236          PUT   HANOIOUT,OMSG       Write error message               00073901
0002D2 4110 C46E      00474         237+         LA    1,HANOIOUT                    LOAD PARAMETER REG 1   01900002
0002D6 4100 C6C9      006CF         238+         LA    0,OMSG                        LOAD PARAMETER REG 0   02500002
0002DA 58F0 1030      00030         239+         L     15,48(0,1)              LOAD PUT ROUTINE ADDR        00550000
0002DE 05EF                         240+         BALR  14,15                   LINK TO PUT ROUTINE          00600000
0002E0 41B0 0008      00008         241          LA    R11,8                 return code                    00074001
0002E4 47F0 C310      00316         242          B     PROCEND               end of program                 00074101
                                    243 *                                                                    00074201
                            002E8   244 ERRP4    EQU   *                                                    00074301
0002E8 D203 C6CD C766 006D3 0076C   245          MVC   OMSGNO,=C'0004'                                       00074401
0002EE D200 C6D1 C81A 006D7 00820   246          MVC   OMSGTYPE,=C'E'                                        00074501
0002F4 D27A C6D3 C644 006D9 0064A   247          MVC   OMSGTEXT,BLANK                                        00074601
0002FA D230 C6D3 C861 006D9 00867   248          MVC   OMSGTEXT(49),=C'Algorithm spcification must be either REX00074701
                                                        C or ITER'                                          00074801
                                    249          PUT   HANOIOUT,OMSG       Write error message               00074901
000300 4110 C46E      00474         250+         LA    1,HANOIOUT                    LOAD PARAMETER REG 1   01900002
000304 4100 C6C9      006CF         251+         LA    0,OMSG                        LOAD PARAMETER REG 0   02500002
000308 58F0 1030      00030         252+         L     15,48(0,1)              LOAD PUT ROUTINE ADDR        00550000
00030C 05EF                         253+         BALR  14,15                   LINK TO PUT ROUTINE          00600000
00030E 41B0 0008      00008         254          LA    R11,8                 return code                    00075001
000312 47F0 C310      00316         255          B     PROCEND               end of program                 00075101
                                    256 * end of program processing                                          00075201
                            00316   257 PROCEND  EQU   *                                                    00075301
                                    258 * close files                                                        00075401
                                    259          CLOSE (HANOIIN)                                             00075501
000316 0700                         260+         CNOP  0,4                     ALIGN LIST TO FULLWORD 02420002
000318 4510 C31A      00320         261+         BAL   1,*+8                   LOAD REG1 W/LIST ADDR  02460002
00031C 80                           262+         DC    AL1(128)                OPTION BYTE            02580000
00031D 000414                       263+         DC    AL3(HANOIIN)            DCB ADDRESS            02600000
000320 0A14                         264+         SVC   20                      ISSUE CLOSE SVC        01640000
                                    265          CLOSE (HANOIOUT)                                            00075601
000322 0700                         266+         CNOP  0,4                     ALIGN LIST TO FULLWORD 02420002
000324 4510 C326      0032C         267+         BAL   1,*+8                   LOAD REG1 W/LIST ADDR  02460002
000328 80                           268+         DC    AL1(128)                OPTION BYTE            02580000
000329 000474                       269+         DC    AL3(HANOIOUT)           DCB ADDRESS            02600000
00032C 0A14                         270+         SVC   20                      ISSUE CLOSE SVC        01640000
                                    271 * handle trailer message to console                                  00075701
00032E 12BB                         272          LTR   R11,R11                                               00075801
000330 4780 C362      00368         273          BE    PROCENDA                                              00075901
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  LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                ASM 0201 20.50 11/01/26

                                    274          WTO   'HANOIASM... Program ended with errors'                  00076001
000334                              275+         CNOP  0,4                                                       09800002
000334 4510 C35C      00362         276+         BAL   1,IHB0034A                    BRANCH AROUND MESSAGE        09850002
000338 0029                         277+         DC    AL2(41)             TEXT LENGTH                          13200002
00033A 0000                         278+         DC    B'0000000000000000' MCS FLAGS                            13250002
00033C C8C1D5D6C9C1E2D4             279+         DC    C'HANOIASM... Program ended with errors'                 13350002
000362                              280+IHB0034A DS    0H                                                       15850002
000362 0A23                         281+         SVC   35                                                       15950002
000364 47F0 C38E      00394         282          B     PROCENDB                                                 00076101
                            00368   283 PROCENDA EQU   *                                                        00076201
                                    284          WTO   'HANOIASM... Normal end of program'                    00076301
000368                              285+         CNOP  0,4                                                       09800002
000368 4510 C38C      00392         286+         BAL   1,IHB0035A                    BRANCH AROUND MESSAGE        09850002
00036C 0025                         287+         DC    AL2(37)             TEXT LENGTH                          13200002
00036E 0000                         288+         DC    B'0000000000000000' MCS FLAGS                            13250002
000370 C8C1D5D6C9C1E2D4             289+         DC    C'HANOIASM... Normal end of program'                     13350002
000392                              290+IHB0035A DS    0H                                                       15850002
000392 0A23                         291+         SVC   35                                                       15950002
                            00394   292 PROCENDB EQU   *                                                        00076401
                                    293 * return to caller                                                     00076501
000394 58D0 C3A2      003A8         294          L     R13,SAVEAREA+4                                           00076601
000398 18FB                         295          LR    R15,R11                                                  00076701
                                    296          RETURN (14,12),RC=(15)                                         00076801
00039A 58ED 000C      0000C         297+         L     14,12(13,0)                   RESTORE REGISTER 14        01200000
00039E 980C D014      00014         298+         LM    0,12,20(13)                   RESTORE THE REGISTERS      01000000
0003A2 07FE                         299+         BR    14                                  RETURN               02000000
                                    300 *--- data definitions                                                  00076901
                                    301 * INSTANCE - same structure as in HANOILN to allow                     00077001
                                    302 * return values using this structure (STEPS)                         00077101
0003A4                              303 SAVEAREA DS    18F                                                      00077201
0003EC 00000474                     304 OFDUMMY  DC    A(HANOIOUT)         address output file DCB - not used   00077301
                                    305 * fields also used as parameters                                       00077401
0003F0                              306 SLICES   DS    PL2                 number of slices to move             00077501
0003F2                              307 FROM     DS    PL1                 from stack number                    00077601
0003F3                              308 TO       DS    PL1                 to stack number                      00077701
0003F4                              309 AUX      DS    PL1                 auxiliary stack number               00077801
0003F5                              310 STEPS    DS    PL4                 number of steps                      00077901
0003F9 404040                       311          DC    CL3'   '                                               00078001
                                    312 * Parameters to HANOILN (proper way to pass parameters...)             00078101
0003FC 00000474                     313 ALIST    DC    A(HANOIOUT)                                              00078201
000400 000003F0                     314          DC    A(SLICES)                                                00078301
000404 000003F2                     315          DC    A(FROM)                                                  00078401
000408 000003F3                     316          DC    A(TO)                                                    00078501
00040C 000003F4                     317          DC    A(AUX)                                                   00078601
000410 80                           318          DC    X'80'                                                    00078701
000411 0003F5                       319          DC    AL3(STEPS)                                               00078801
                                    320 *--- file definitions                                                  00078921
                                    321 HANOIIN  DCB   DSORG=PS,RECFM=FB,LRECL=80,MACRF=(GM),EODAD=ATFIEND,    X00079001
                                                      DEVD=DA,DDNAME=HANOIIN                                    00079101


                                    323+*                     DATA CONTROL BLOCK                           22770000
                                    324+*                                                                  22860000
000414                              325+HANOIIN  DC    0F'0'                ORIGIN ON WORD BOUNDARY          22914000
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 LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                ASM 0201 20.50 11/01/26

                                   327+*                    DIRECT ACCESS DEVICE INTERFACE                   27360000

000414 0000000000000000            329+         DC    BL16'0'                  FDAD,DVTBL                     27540000
000424 00000000                    330+         DC    A(0)                     KEYLE,DEVT,TRBAL               27720000

                                   332+*                    COMMON ACCESS METHOD INTERFACE                   48690000

000428 00                          334+         DC    AL1(0)                   BUFNO                          49050000
000429 000001                      335+         DC    AL3(1)                   BUFCB                          54720000
00042C 0000                        336+         DC    AL2(0)              BUFL                                55170000
00042E 4000                        337+         DC    BL2'0100000000000000'                                  *55800000
                                     +                                                 DSORG                  55890000
000430 00000001                    338+         DC    A(1)                     IOBAD                          56340000

                                   340+*                    FOUNDATION EXTENSION                             56610000

000434 00                          342+         DC    BL1'00000000'                         BFTEK,BFLN,HIARCHY 59850000
000435 00025E                      343+         DC    AL3(ATFIEND)             EODAD                          65970000
000438 90                          344+         DC    BL1'10010000'                                           *66150000
                                     +                                        RECFM                          66240000
000439 000000                      345+         DC    AL3(0)                   EXLST                          66330000

                                   347+*                    FOUNDATION BLOCK                                 66690000

00043C C8C1D5D6C9C9D540            349+         DC    CL8'HANOIIN'             DDNAME                         66870000
000444 02                          350+         DC    BL1'00000010'            OFLGS                          68220000
000445 00                          351+         DC    BL1'00000000'                              IFLG         68310000
000446 5000                        352+         DC    BL2'0101000000000000'                                  *68400000
                                     +                                                                       *68490000
                                     +                                        MACR                           68580000

                                   354+*                    BSAM-BPAM-QSAM INTERFACE                         74430000

000448 00                          356+         DC    BL1'00000000'                                           *74610000
                                     +                                                                 RER1 74700000
000449 000001                      357+         DC    AL3(1)                   CHECK, GERR, PERR              74790000
00044C 00000001                    358+         DC    A(1)                     SYNAD                          74880000
000450 0000                        359+         DC    H'0'                     CIND1, CIND2                   74970000
000452 0000                        360+         DC    AL2(0)                   BLKSIZE                        75240000
000454 00000000                    361+         DC    F'0'                     WCPO, WCPL, OFFSR, OFFSW       75870000
000458 00000001                    362+         DC    A(1)                     IOBA                           75960000
00045C 00                          363+         DC    AL1(0)                   NCP                            76050000
00045D 000001                      364+         DC    AL3(1)                   EOBR, EOBAD                    76140000

                                   366+*                    QSAM INTERFACE                                   81450000

000460 00000001                    368+         DC    A(1)                      RECAD                         81630000
000464 0000                        369+         DC    H'0'                     QSWS                           81810000
000466 0050                        370+         DC    AL2(80)           LRECL                                80730000
000468 00                          371+         DC    BL1'00000000'            EROPT                          82530000
000469 000001                      372+         DC    AL3(1)                   CNTRL                          82620000
00046C 00000000                    373+         DC    F'0'                     PRECL                          82710000
000470 00000001                    374+         DC    A(1)                     EOB                            82800000
                                   375 HANOIOUT DCB   DSORG=PS,RECFM=FBA,LRECL=133,MACRF=(PM),               X00079201
                                                      DEVD=DA,DDNAME=HANOIOUT                                00079301
```

```
 LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                ASM 0201 20.50 11/01/26

                                   377+*                    DATA CONTROL BLOCK                                    22770000
                                   378+*                                                                          22860000
000474                             379+HANOIOUT DC    0F'0'                  ORIGIN ON WORD BOUNDARY              22914000

                                   381+*                    DIRECT ACCESS DEVICE INTERFACE                        27360000

000474 0000000000000000            383+         DC    BL16'0'                 FDAD,DVTBL                           27540000
000484 00000000                    384+         DC    A(0)                    KEYLE,DEVT,TRBAL                     27720000

                                   386+*                    COMMON ACCESS METHOD INTERFACE                        48690000

000488 00                          388+         DC    AL1(0)                  BUFNO                                49050000
000489 000001                      389+         DC    AL3(1)                  BUFCB                                54720000
00048C 0000                        390+         DC    AL2(0)             BUFL                                      55170000
00048E 4000                        391+         DC    BL2'0100000000000000'                                       *55800000
                                     +                                                 DSORG                      55890000
000490 00000001                    392+         DC    A(1)                    IOBAD                                56340000

                                   394+*                    FOUNDATION EXTENSION                                  56610000

000494 00                          396+         DC    BL1'00000000'                BFTEK,BFLN,HIARCHY              59850000
000495 000001                      397+         DC    AL3(1)                  EODAD                                65970000
000498 94                          398+         DC    BL1'10010100'                                               *66150000
                                     +                                       RECFM                              66240000
000499 000000                      399+         DC    AL3(0)                  EXLST                                66330000

                                   401+*                    FOUNDATION BLOCK                                      66690000

00049C C8C1D5D6C9D6E4E3            403+         DC    CL8'HANOIOUT'           DDNAME                               66870000
0004A4 02                          404+         DC    BL1'00000010'           OFLGS                                68220000
0004A5 00                          405+         DC    BL1'00000000'                        IFLG                    68310000
0004A6 0050                        406+         DC    BL2'0000000001010000'                                       *68400000
                                     +                                                                            *68490000
                                     +                                       MACR                               68580000

                                   408+*                    BSAM-BPAM-QSAM INTERFACE                              74430000

0004A8 00                          410+         DC    BL1'00000000'                                               *74610000
                                     +                                                                      RER1 74700000
0004A9 000001                      411+         DC    AL3(1)                  CHECK, GERR, PERR                    74790000
0004AC 00000001                    412+         DC    A(1)                    SYNAD                                74880000
0004B0 0000                        413+         DC    H'0'                    CIND1, CIND2                         74970000
0004B2 0000                        414+         DC    AL2(0)                  BLKSIZE                              75240000
0004B4 00000000                    415+         DC    F'0'                    WCPO, WCPL, OFFSR, OFFSW             75870000
0004B8 00000001                    416+         DC    A(1)                    IOBA                                 75960000
0004BC 00                          417+         DC    AL1(0)                  NCP                                  76050000
0004BD 000001                      418+         DC    AL3(1)                  EOBR, EOBAD                          76140000

                                   420+*                    QSAM INTERFACE                                        81450000

0004C0 00000001                    422+         DC    A(1)                     RECAD                               81630000
0004C4 0000                        423+         DC    H'0'                    QSWS                                 81810000
0004C6 0085                        424+         DC    AL2(133)       LRECL                                     80730000
0004C8 00                          425+         DC    BL1'00000000'           EROPT                                82530000
0004C9 000001                      426+         DC    AL3(1)                  CNTRL                                82620000
```

```
  LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                ASM 0201 20.50 11/01/26

0004CC 00000000                    427+         DC    F'0'                  PRECL                          82710000
0004D0 00000001                    428+         DC    A(1)                  EOB                            82800000
                                   429 *--- input record definition                                          00079421
0004D4                             430 IREC     DS    0CL80              input record                        00079501
0004D4                             431          DS    CL1' '             comment marker (ASM editor profile) 00079601
0004D5                             432 ISLICES  DS    CL2                no of slices (0<=SLICES<=11)        00079701
0004D7                             433          DS    CL1' '             filler                              00079801
0004D8                             434 IFROM    DS    CL1                from stack (1<=FROM<=3)             00079901
0004D9                             435          DS    CL1' '             filler                              00080001
0004DA                             436 ITO      DS    CL1                to stack (1<=TO<=3,TO<>FROM)        00080101
0004DB                             437          DS    CL1' '             filler                              00080201
0004DC                             438 IALG     DS    CL4                algorithm REC or ITER               00080301
0004E0                             439          DS    CL68               filler                              00080401
                                   440 *--- output record definitions                                        00080521
                                   441 * output record definition for headline                               00080621
000524                             442 OHEAD    DS    0CL133                                                 00080701
000524 40                          443          DC    CL1' '             ASO character                       00080801
000525 D496A589958740              444          DC    CL7'Moving '                                           00080901
00052C                             445 OHSLICES DS    CL2                                                    00081001
00052E 40A293898385A240            446          DC    CL26' slices from source stack '                       00081101
000548                             447 OHFROM   DS    CL1                                                    00081201
000549 40A396408485A2A3            448          DC    CL22' to destination stack '                           00081301
00055F                             449 OHTO     DS    CL1                                                    00081401
000560 40A4A28995874081            450          DC    CL23' using auxiliary stack '                          00081501
000577                             451 OHAUX    DS    CL1                                                    00081601
000578 4040404040404040            452          DC    CL49' '                                                00081701
                                   453 * output record definition for trailing line                          00081801
0005A9                             454 OTRAIL   DS    0CL133                                                 00081901
0005A9 40                          455          DC    CL1' '             ASO character                       00082001
0005AA C496958540899540            456          DC    CL8'Done in '                                          00082101
0005B2                             457 OTSTEPS  DS    CL7                                                    00082201
0005B9 40A2A38597A2                458          DC    CL6' steps'                                            00082301
0005BF 4040404040404040            459          DC    CL111' '                                               00082401
                                   460 * templates and target fields for print preparation                   00082501
00062E                             461 TRGTSLC  DS    CL4                                                    00082601
000632 40202120                    462 TMPSLC   DC    X'40202120'                                            00082719
000636                             463 TRGTSTK  DS    CL2                                                    00082801
000638 4020                        464 TMPSTK   DC    X'4020'                                                00082919
00063A                             465 TRGTSTPS DS    CL8                                                    00083001
000642 4020202020202120            466 TMPSTPS  DC    X'4020202020202120'                                    00083119
                                   467 * output record definition for error messages                        00083201
00064A 4040404040404040            468 BLANK    DC    133CL1' '          all blank                           00083301
0006CF                             469 OMSG     DS    0CL133                                                 00083401
0006CF 40                          470          DC    CL1' '             ASO character                       00083501
0006D0 C8C1D5                      471          DC    CL3'HAN'           message no prefix                   00083601
0006D3                             472 OMSGNO   DS    CL4                message no                          00083701
0006D7                             473 OMSGTYPE DS    CL1                message type (I,W,E)                00083801
0006D8 40                          474          DC    CL1' '             filler                              00083901
0006D9                             475 OMSGTEXT DS    CL123              message text                        00084001
                                   476 *--- Literals, if any, will go here                                   00084101
000758                             477          LTORG *                                                      00084201
000758 D9C5C340                    478                =C'REC '
00075C C9E3C5D9                    479                =C'ITER'
000760 F0F0F0F1                    480                =C'0001'
000764 F0F0F0F2                    481                =C'0002'
```

```
  LOC  OBJECT CODE    ADDR1 ADDR2  STMT   SOURCE STATEMENT                                ASM 0201 20.50 11/01/26

000768 F0F0F0F3                      482               =C'0003'
00076C F0F0F0F4                      483               =C'0004'
000770 020C                          484               =P'20'
000772 D5A4948285994096              485               =C'Number of slices must be 1<=slices<=20'
000798 0C                            486               =P'0'
000799 3C                            487               =P'3'
00079A 6C                            488               =P'6'
00079B 4040404040404040              489               =CL133' '
000820 C5                            490               =C'E'
000821 D59640899597A4A3              491               =C'No input record found'
000836 E58193A485A24086              492               =C'Values from,to must be 1<=from,to<=3 and from<>to'
000867 C19387969989A388              493               =C'Algorithm spcification must be either REC or ITER'
                                     494 *--- end of section                                                  00084301
000000                               495          END   HANOIASM                                              00084401
```

POS.ID	REL.ID	FLAGS	ADDRESS
0002	0002	08	000045
0002	0002	08	000051
0002	0002	08	00018D
0002	0002	08	0001F9
0002	0002	08	00031D
0002	0002	08	000329
0002	0002	0C	0003EC
0002	0002	0C	0003FC
0002	0002	0C	000400
0002	0002	0C	000404
0002	0002	0C	000408
0002	0002	0C	00040C
0002	0002	08	000411
0002	0002	08	000435
0002	0003	1C	000184
0002	0004	1C	0001F0

SYMBOL	LEN	VALUE	DEFN	REFERENCES
ALIST	00004	000003FC	00313	00146 00176
ATFIEND	00001	0000025E	00205	00343
AUX	00001	000003F4	00309	00099 00100 00101 00113 00317
BLANK	00001	0000064A	00468	00208 00221 00234 00247
CALLREC	00001	000001C4	00159	00129
C4OK	00001	000000D0	00097	00094
ERRP1	00001	0000028C	00218	00075 00077
ERRP23	00001	000002BA	00231	00081 00083 00087 00089 00091
ERRP4	00001	000002E8	00244	00096
FROM	00001	000003F2	00307	00079 00080 00082 00090 00100 00107 00315
HANOIASM	00001	00000000	00034	00495
HANOIIN	00004	00000414	00325	00058 00068 00263
HANOIOUT	00004	00000474	00379	00064 00116 00121 00194 00199 00211 00224 00237 00250 00269 00304 00313
IALG	00004	000004DC	00438	00093 00095 00128
IFROM	00001	000004D8	00434	00079
IHB0003A	00002	00000038	00051	00047
IHB0012A	00002	0000017C	00136	00132
IHB0013B	00004	00000184	00141	00148
IHB0014A	00001	00000190	00147	00143
IHB0015A	00002	000001BE	00156	00152
IHB0016A	00002	000001E8	00166	00162
IHB0017B	00004	000001F0	00171	00178
IHB0018A	00001	000001FC	00177	00173
IHB0019A	00002	0000022A	00186	00182
IHB0034A	00002	00000362	00280	00276
IHB0035A	00002	00000392	00290	00286
IREC	00080	000004D4	00430	00069
ISLICES	00002	000004D5	00432	00073
ITO	00001	000004DA	00436	00085
OHAUX	00001	00000577	00451	00114
OHEAD	00133	00000524	00442	00117
OHFROM	00001	00000548	00447	00108
OHSLICES	00002	0000052C	00445	00105
OHTO	00001	0000055F	00449	00111
OMSG	00133	000006CF	00469	00212 00225 00238 00251
OMSGNO	00004	000006D3	00472	00206 00219 00232 00245
OMSGTEXT	00123	000006D9	00475	00208 00209 00221 00222 00234 00235 00247 00248
OMSGTYPE	00001	000006D7	00473	00207 00220 00233 00246
OTRAIL	00133	000005A9	00454	00200
OTSTEPS	00007	000005B2	00457	00192
PREPEND	00001	0000022C	00189	00158
PROCEND	00001	00000316	00257	00203 00216 00229 00242 00255
PROCENDA	00001	00000368	00283	00273
PROCENDB	00001	00000394	00292	00282
R11	00001	0000000B	00029	00053 00215 00228 00241 00254 00272 00272 00295
R12	00001	0000000C	00030	00038 00039
R13	00001	0000000D	00031	00041 00042 00043 00294
R15	00001	0000000F	00033	00295
R2	00001	00000002	00020	00040 00041 00043
SAVEAREA	00004	000003A4	00303	00040 00042 00294
SLICES	00002	000003F0	00306	00073 00074 00076 00104 00314
STEPS	00004	000003F5	00310	00126 00191 00319
TMPSLC	00004	00000632	00462	00103
TMPSTK	00002	00000638	00464	00106 00109 00112
TMPSTPS	00008	00000642	00466	00190

```
SYMBOL    LEN   VALUE   DEFN    REFERENCES

TO       00001 000003F3 00308  00085 00086 00088 00090 00101 00110 00316
TRGTSLC  00004 0000062E 00461  00103 00104 00105
TRGTSTK  00002 00000636 00463  00106 00107 00108 00109 00110 00111 00112 00113 00114
TRGTSTPS 00008 0000063A 00465  00190 00191 00192
```

```
SYMBOL    LEN   VALUE   DEFN    REFERENCES

=C'REC ' 00004 00000758 00478  00093 00128
=C'ITER' 00004 0000075C 00479  00095
=C'0001' 00004 00000760 00480  00206
=C'0002' 00004 00000764 00481  00219
=C'0003' 00004 00000768 00482  00232
=C'0004' 00004 0000076C 00483  00245
=P'20'   00002 00000770 00484  00076
=C'Number of slices must be 1<=slices<=20'
         00038 00000772 00485  00222
=P'0'    00001 00000798 00486  00074 00080 00086 00126
=P'3'    00001 00000799 00487  00082 00088
=P'6'    00001 0000079A 00488  00099
=CL133' '
         00133 0000079B 00489  00122 00195
=C'E'    00001 00000820 00490  00207 00220 00233 00246
=C'No input record found'
         00021 00000821 00491  00209
=C'Values from,to must be 1<=from,to<=3 and from<>to'
         00049 00000836 00492  00235
=C'Algorithm spcification must be either REC or ITER'
         00049 00000867 00493  00248
```

```
NO STATEMENTS FLAGGED IN THIS ASSEMBLY
HIGHEST SEVERITY WAS    0
OPTIONS FOR THIS ASSEMBLY
  ALIGN, ALOGIC, BUFSIZE(STD), NODECK, ESD, FLAG(0), LINECOUNT(55), LIST, NOMCALL, YFLAG, WORKSIZE(2097152)
  NOMLOGIC, NONUMBER, OBJECT, NORENT, RLD, NOSTMT, NOLIBMAC, NOTERMINAL, NOTEST, XREF(SHORT)
  SYSPARM()
WORK FILE BUFFER SIZE/NUMBER = 8702/ 1
TOTAL RECORDS READ FROM SYSTEM INPUT        229
TOTAL RECORDS READ FROM SYSTEM LIBRARY     3902
TOTAL RECORDS PUNCHED                        49
TOTAL RECORDS PRINTED                       652
```

```
 F64-LEVEL LINKAGE EDITOR OPTIONS SPECIFIED XREF,LET,LIST,NCAL
         DEFAULT OPTION(S) USED -  SIZE=(98304,55296)
IEW0000                  ENTRY HANOIASM                                          00084801



                                        CROSS REFERENCE TABLE


  CONTROL SECTION                       ENTRY

    NAME    ORIGIN  LENGTH                NAME   LOCATION     NAME   LOCATION     NAME   LOCATION     NAME   LOCATION
  HANOIREC      00     2A9
  HANOIITR     2B0     50C
  HANOIASM     7C0     898



  LOCATION  REFERS TO SYMBOL  IN CONTROL SECTION             LOCATION  REFERS TO SYMBOL  IN CONTROL SECTION
      944            HANOIITR        HANOIITR                   9B0            HANOIREC        HANOIREC

 ENTRY ADDRESS      7C0

 TOTAL LENGTH      1058
****GO        DOES NOT EXIST BUT HAS BEEN ADDED TO DATA SET
AUTHORIZATION CODE IS         0.
```

```
Moving  5 slices from source stack 1 to destination stack 2 using auxiliary stack 3

Step       1: Moving slice  1 from stack 1 to 2
Step       2: Moving slice  2 from stack 1 to 3
Step       3: Moving slice  1 from stack 2 to 3
Step       4: Moving slice  3 from stack 1 to 2
Step       5: Moving slice  1 from stack 3 to 1
Step       6: Moving slice  2 from stack 3 to 2
Step       7: Moving slice  1 from stack 1 to 2
Step       8: Moving slice  4 from stack 1 to 3
Step       9: Moving slice  1 from stack 2 to 3
Step      10: Moving slice  2 from stack 2 to 1
Step      11: Moving slice  1 from stack 3 to 1
Step      12: Moving slice  3 from stack 2 to 3
Step      13: Moving slice  1 from stack 1 to 2
Step      14: Moving slice  2 from stack 1 to 3
Step      15: Moving slice  1 from stack 2 to 3
Step      16: Moving slice  5 from stack 1 to 2
Step      17: Moving slice  1 from stack 3 to 1
Step      18: Moving slice  2 from stack 3 to 2
Step      19: Moving slice  1 from stack 1 to 2
Step      20: Moving slice  3 from stack 3 to 1
Step      21: Moving slice  1 from stack 2 to 3
Step      22: Moving slice  2 from stack 2 to 1
Step      23: Moving slice  1 from stack 3 to 1
Step      24: Moving slice  4 from stack 3 to 2
Step      25: Moving slice  1 from stack 1 to 2
Step      26: Moving slice  2 from stack 1 to 3
Step      27: Moving slice  1 from stack 2 to 3
Step      28: Moving slice  3 from stack 1 to 2
Step      29: Moving slice  1 from stack 3 to 1
Step      30: Moving slice  2 from stack 3 to 2
Step      31: Moving slice  1 from stack 1 to 2

Done in      31 steps
```

```
                HH        HH   AAAAAAAAAA   NN        NN  OOOOOOOOOOOO   IIIIIIIIII    AAAAAAAAAA    SSSSSSSSSS   MM        MM
               HH        HH  AAAAAAAAAAAA  NNN       NN  OOOOOOOOOOOO   IIIIIIIIII   AAAAAAAAAAAA  SSSSSSSSSSSS  MMM      MMM
              HH        HH  AA        AA  NNNN      NN  OO        OO       II       AA        AA  SS        SS  MMMM    MMMM
             HH        HH  AA        AA  NN NN     NN  OO        OO       II       AA        AA  SS            MM MM  MM MM
            HH        HH  AA        AA  NN  NN    NN  OO        OO       II       AA        AA  SSS           MM  MMMM  MM
           HHHHHHHHHHHH  AAAAAAAAAAAA  NN   NN   NN  OO        OO       II       AAAAAAAAAAAA   SSSSSSSS    MM   MM   MM
          HHHHHHHHHHHH  AAAAAAAAAAAA  NN    NN  NN  OO        OO       II       AAAAAAAAAAAA    SSSSSSSS   MM        MM
         HH        HH  AA        AA  NN     NN NN  OO        OO       II       AA        AA            SSS  MM        MM
        HH        HH  AA        AA  NN      NNNN  OO        OO       II       AA        AA            SS  MM        MM
       HH        HH  AA        AA  NN       NNN  OO        OO       II       AA        AA  SS        SS  MM        MM
      HH        HH  AA        AA  NN        NN  OOOOOOOOOOOO   IIIIIIIIII   AA        AA  SSSSSSSSSSSS  MM        MM
     HH        HH  AA        AA  NN         N  OOOOOOOOOOOO   IIIIIIIIII   AA        AA   SSSSSSSSSS   MM        MM


                  JJJJJJJJJJ   9999999999   555555555555  555555555555                          AAAAAAAAAA
                  JJJJJJJJJJ  999999999999  555555555555  555555555555                         AAAAAAAAAAAA
                      JJ      99        99  55            55                                   AA        AA
                      JJ      99        99  55            55                                   AA        AA
                      JJ      99        99  55            55                                   AA        AA
                      JJ      999999999999  555555555     555555555                            AAAAAAAAAAAA
                      JJ      999999999999  5555555555    5555555555                           AAAAAAAAAAAA
                      JJ                99          55            55                           AA        AA
                JJ    JJ                99           55            55                          AA        AA
                JJ    JJ      99        99           55            55                          AA        AA
                JJJJJJJ       999999999999  555555555555  555555555555                        AA        AA
                 JJJJJJ        9999999999   55555555555   55555555555                         AA        AA


****A   END   JOB  955  HANOIASM  Hanoi ASM            ROOM       8.50.50 PM 01 NOV 26  PRINTER1  SYS TK4-  JOB  955   END   A****
****A   END   JOB  955  HANOIASM  Hanoi ASM            ROOM       8.50.50 PM 01 NOV 26  PRINTER1  SYS TK4-  JOB  955   END   A****
****A   END   JOB  955  HANOIASM  Hanoi ASM            ROOM       8.50.50 PM 01 NOV 26  PRINTER1  SYS TK4-  JOB  955   END   A****
```